



**Verified Carbon
Standard**

UMOYILANGA HYBRID PROJECT

Anthesis 

umoyilanga

Project title	Umoyilanga Hybrid Project
Project ID	5272
Crediting period	15-October-2025 to 14-October-2032
Original date of issue	25 October 2024
Most recent date of issue	10 June 2025
Version	2.3.2
VCS Standard Version	4.7
Prepared by	Grant Little - Anthesis South Africa & Carbon Development Team

CONTENTS

1	PROJECT DETAILS.....	4
1.1	Summary Description of the Project	4
1.2	Audit History.....	5
1.3	Sectoral Scope and Project Type	5
1.4	Project Eligibility	5
1.5	Project Design	8
1.6	Project Proponent	9
1.7	Other Entities Involved in the Project	9
1.8	Ownership.....	9
1.9	Project Start Date	10
1.10	Project Crediting Period	10
1.11	Project Scale and Estimated GHG Emission Reductions or Removals	11
1.12	Description of the Project Activity	11
1.13	Project Location	16
1.14	Conditions Prior to Project Initiation	16
1.15	Compliance with Laws, Statutes and Other Regulatory Frameworks	18
1.16	Double Counting and Participation under Other GHG Programs	21
1.17	Double Claiming, Other Forms of Credit, and Scope 3 Emissions.....	21
1.18	Sustainable Development Contributions	22
1.19	Additional Information Relevant to the Project	25
2	SAFEGUARDS AND STAKEHOLDER ENGAGEMENT	26
2.1	Stakeholder Engagement and Consultation.....	26
2.2	Risks to Stakeholders and the Environment.....	35
2.3	Respect for Human Rights and Equity	37
2.4	Ecosystem Health	39
3	APPLICATION OF METHODOLOGY.....	46
3.1	Title and Reference of Methodology	46
3.2	Applicability of Methodology	46
3.3	Project Boundary	55
3.4	Baseline Scenario	57

3.5	Additionality	59
3.6	Methodology Deviations	66
4	QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS.....	67
4.1	Baseline Emissions	67
4.2	Project Emissions	79
4.3	Leakage Emissions	80
4.4	Estimated GHG Emission Reductions and Carbon Dioxide Removals	80
5	MONITORING	82
5.1	Data and Parameters Available at Validation	82
5.2	Data and Parameters Monitored	97
5.3	Monitoring Plan	100
	APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION.....	103
	APPENDIX 2: BIBLIOGRAPHY	104

1 PROJECT DETAILS

1.1 Summary Description of the Project

The project combines solar, wind and battery storage technologies to offer dispatchable and reliable power to the national electricity grid. The project will operate as a virtual power plant, combining generation from two sites which are located over 700 km apart, namely Avondale in the Northern Cape, with 115 MW of solar PV and 30 MW of battery storage, and Dassiesridge in the Eastern Cape, with 63 MW of wind and 45 MW of battery storage.

The unique combination of wind and solar resources with battery storage enables Umoyilanga to provide 75 MW dispatchable electricity on demand from 05:00 to 21:30 daily as per the requirements of the PPA, and demonstrates that renewable energy can provide reliable, dispatchable power at a competitive price.

To achieve this, batteries at Dassiesridge will generally charge from the wind energy at night, discharging power in the morning until the sun rises. The solar installation at Avondale will supply the bulk of the energy during the day, supplemented by wind energy from Dassiesridge. Excess solar energy will be used to charge the batteries at Avondale, which will be discharged after sunset. A sophisticated virtual power plant controller (VPP) energy management system will give instructions to assets across both sites to optimise the power supply in real time, depending on weather forecasts and Eskom's requirements. Depending on the availability of renewable energy at each time increment, the hybrid virtual power plant controller will optimally meet the contracted energy demand requirements of the PPA either directly from the solar PV or wind, from the BESS, or from a combination of these. Excess power is used to charge the battery energy storage system (BESS) to supply dispatchable power to the grid for prolonged periods as specified in the PPA.

The EMS is composed of one VPP Optimizer & Controller (the VPPOC) which operates on the cloud and calculates the appropriate setpoints for each Power Plant and a local HPPC (Hybrid Power Plant Controller) installed on each power plant which actively controls the assets on each site in real time. The VPPOC maximises the revenues considering plant availability, resource forecast, BESS State of Health etc. Essentially it computes: The capacity to be declare (day-ahead) or redeclare (intraday). The set points for each site to respond to the dispatch instruction to be followed by the HPPC, including Forecasted solicitation of the BESS & Level of generation of the PV and Wind. The low-carbon electricity produced will help to meet the electricity needs of 120,000 households for 20 years, based on the residential consumption average of 3,319 kWh/household annually [1].

Scenario existing prior to implementation of project activity

The electricity delivered to the grid by the project activity would have been generated by the operation of fossil fuel-based power plants. The project activity generates renewable electricity and helps to displace CO₂ emission from fossil fuel-based grid connected power plants.

The estimated annual average CO₂ emission reduction from the project activity is 369,232 tCO₂e/annum. The total CO₂ emission reduction for the first crediting period is estimated at 2,584,626 tCO₂e.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
<i>Validation</i>	15-October-2025 to 14-October-2032	VCS	Carbon Check (India) Private Ltd.	7 (Seven)

1.3 Sectoral Scope and Project Type

Sectoral scope¹	01 - Energy industries (Renewable/Non-Renewable sources)
Project activity type	Renewable Energy Project

1.4 Project Eligibility

1.4.1 General eligibility

As per paragraph 2.1.1 of the VCS Standard, version 4.7, the project is eligible under the scope of the VCS Program because this project activity meets three of the listed criteria, as described in Table 1 below. The project is also eligible under VCS v4.7, table 2.1, as shown in Table 2 below:

Table 1: Scope of the VCS program

Elements	Justification of project applicability
1) The seven Kyoto Protocol greenhouse gases.	The project aims at reducing greenhouse gas (GHG) emissions by replacing fossil fuel-based energy supplied to the South African national grid, with clean renewable electricity.
2) Ozone-depleting substances (ODS).	Not applicable
3) Project activities supported by a methodology approved under the VCS Program through the methodology development and review process.	Project activity applies VCS approved CDM methodology ACM0002: Grid-connected electricity generation from renewable sources

¹ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

	(Version 22.0) ² which has been approved for use under the VCS Program.
4) Project activities supported by a methodology approved under an approved GHG program, unless explicitly excluded (see the Verra website for exclusions).	Project activity applies VCS approved CDM methodology ACM0002: Grid-connected electricity generation from renewable sources (Version 22.0) ³ .
5) Jurisdictional REDD+ programs and nested REDD+ projects as set out in the Jurisdictional and Nested REDD+ (JNR) Requirements.	N/A

Table 2: Excluded Project Activities under the VCS v 4.7 table 1

Activity	Applicability
Grid-connected electricity generation activities using hydroelectric power plants. The exclusion does not apply to ocean energy (e.g., wave, tidal, salinity gradient, and ocean thermal energy conversion). For hydro projects, large scale means a maximum capacity of greater than 15MW, where maximum capacity is the installed/rated capacity or authorized capacity (as determined in the activity approval from the project regulator, government, or similar entity), whichever is lower. Grid-connected means >50% of total generation is exported to a national or regional grid.	Not applicable to this project which is not a hydroelectric power plant.
Grid-connected electricity generation activities using wind, geothermal, or solar photovoltaic (PV) power plants. The exclusion does not apply to concentrated solar thermal-to-electricity, floating solar PV, or energy storage systems (e.g., batteries). Grid-connected means >50% of total generation is exported to a national or regional grid. See the VCS Program Definitions for the full definition.	This is a grid connected electricity generation activity using a combination of wind and solar photovoltaic (PV) but the exclusion does not apply to the energy storage system (BESS) with a hybrid power

² <https://cdm.unfccc.int/methodologies/DB/XB1TX7TAZ6SLWM9B7BC67THHVD16JV>
³ <https://cdm.unfccc.int/methodologies/DB/XB1TX7TAZ6SLWM9B7BC67THHVD16JV>

	controller which feeds to the grid. Hence the project is eligible and not excluded.
Activities recovering waste heat for combined cycle electricity generation, or to heat/cool via cogeneration or trigeneration. The exclusion does not apply to waste gas recovery or electricity generation using waste heat recovery outside of combined cycle applications (e.g., organic Rankine cycles).	Not applicable to this project which is not a waste heat recovery project.
Activities generating electricity and/or thermal energy for industrial use from the combustion of non-renewable biomass, agro-residue biomass, or forest residue biomass. The exclusion does not apply to gasification, pyrolysis, combusting biofuels, biogas, fractions of renewable biomass in refuse-derived fuels, agro/forest biomass residues in waste streams that are sent to landfills, CO₂ capture and storage from renewable biomass combustion, or thermal efficiency improvements (e.g., cook stoves).	Not applicable to this project which does combust biomass.
Activities generating electricity and/or thermal energy using fossil fuels, and activities that involve switching from a higher to a lower carbon content fossil fuel. The exclusion does not apply to the use of captured flare and/or vent gas or waste containing previously used petroleum products (e.g., used plastics, oils, lubricants).	Not applicable to this project which does not combust fossil fuel.
Activities replacing electric lighting with more energy-efficient electric lighting, such as the replacement of incandescent electrical bulbs with compact fluorescent lights (CFLs) or light emitting diodes (LEDs). Large-scale means energy-efficient improvements with a maximum savings greater than 60 GWh/year or emission reduction greater than 60 kt CO₂e per year.	Not applicable to this project which does not replace electric lighting.
Activities installing and/or replacing electricity transmission lines and/or energy-efficient transformers. Large-scale means energy-efficient improvements with a maximum savings greater than 60 GWh/year or emission reduction greater than 60 kt CO₂e per year.	Not applicable to this project which does not install or replace electricity transmission

	lines or transformers.
Activities that reduce hydrofluorocarbon-23 (HFC-23) emissions	Not applicable to this project which does not reduce HFC-23 emissions.

Pipeline listing and validation deadlines - The project activity was listed on the VCS Registry before the validation process began. The opening meeting with the VVB took place after the project was listed on the project pipeline with the status of 'Under Validation'. Furthermore, validation did not begin until the 30-day public comment period had begun, and the VVB did not complete validation until after the 30-day public comment period had ended. As per para 3.8.1 of the VCS standard V4.7, non-AFOLU projects shall complete validation within two years of the project start date and this condition will be fulfilled by this project activity as the start date of the project activity is anticipated to be 15 October 2025 (being the day that planned generation will be fed into the national grid and national utility sign-off is achieved) and validation commenced in 2024.

Project activity applies the VCS approved CDM methodology ACM0002: Grid-connected electricity generation from renewable sources (Version 22.0), which has been approved by Verra for use under the VCS Programme for project activities seeking registration with them. ACM0002 is a large-scale methodology and has no capacity limit, and the Umoyilanga Hybrid project is a single project with multiple locations and project activities and is not part of a larger project or activity.

1.4.2 AFOLU project eligibility

This section is not applicable as the project is not an AFOLU project.

1.4.3 Transfer project eligibility

This section is not applicable as the project is not being transferred from another GHG Program nor a CPA seeking registration under the VCS Program.

1.5 Project Design

Project has been designed as:

- ☐ Single location or installation
- ☒ Multiple locations or project activity instances (but not a grouped project)
- ☐ Grouped project

1.5.1 Grouped project design

This section is not applicable as the Project is not a Grouped project, but a single project with multiple project activity instances at multiple locations. The design of the multiple project activities is described under section 1,12 in this document.

1.6 Project Proponent

Organization name	Anthesis B.V.
Contact person	Grant Little
Title	Principal Carbon Developer
Address	Workshop 17, 32 Kloof Street, Cape Town, South Africa
Telephone	+27 21 300 6770
Email	grant.little@anthesisgroup.com

Organization name	Umoyilanga (RF) (Pty) Ltd
Contact person	Gregoire de Montgolfier
Title	Project Director
Address	1st Floor Fernwood House, the Oval – 1 Oakdale Road, Newlands, Cape Town, Western Cape, 7800
Telephone	+27 (0)63 638 9763
Email	gregoire.demontgolfier@edf-re.co.za

1.7 Other Entities Involved in the Project

There are no other entities involved in the development of this project.

1.8 Ownership

Umoyilanga (RF) (Pty) Ltd has the legal right to own and operate the Umoyilanga Hybrid facilities, which can be verified through the following documents provided by the project proponent:

- Eskom's Power Purchase Agreement dated 30 August 2023 which includes all details of the physical locations and property boundaries [2].
- NERSA Generation Licence dated 21 September 2021 [3].
- The Operator has a legal right to use the land with land lease agreements for the duration of the project lifetime, which can be verified through the following documents provided by the project proponent:
 - o Avondale Lease Agreement
 - o Dassiesridge lease Agreement

Anthesis and Umoyilanga have a legal agreement⁴ in place for the joint ownership of a registered carbon project.

1.9 Project Start Date

Project start date	15 October 2025
Justification	According to Section 3.8 in the VCS Standard Version 4.7, the project start date of a non-AFOLU project is the date on which the project begins generating GHG emission reductions or carbon dioxide removals. In this case the start date of project activity is the Commercial Operation Date (COD), on which the plant first starts to feed electricity into the national grid. As such, that is the day on which the project starts generating emission reduction credits. This has been taken as per the Power Purchase Agreement as the date above, when COD is planned. An attestation has been presented as evidence for this.

1.10 Project Crediting Period

Crediting period	☒ Seven years, twice renewable ☐ Ten years, fixed
------------------	---

⁴ Anthesis, Umoyilanga RF Term sheet Commercial Agreement, signed 22 July 2024

	☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of first or fixed crediting period	15 October-2025 to 14-October-2032

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals (ERRs) of the project are:

- ☐ < 300,000 tCO₂e/year (project)
- ☒ ≥ 300,000 tCO₂e/year (large project)

The table below shows the estimated annual GHG emission reductions for the first renewable crediting period:

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
15-October-2025 to 31-December-2025	76 923
01-January-2026 to 31-December-2026	369,232
01-January-2027 to 31-December-2027	369,232
01-January-2028 to 31-December-2028	369,232
01-January-2029 to 31-December-2029	369,232
01-January-2030 to 31-December-2030	369,232
01-January-2031 to 31-December-2031	369,232
01-January-2032 to 14-October-2032	292 309
Total estimated ERRs during the first or fixed crediting period	2,584,626
Total number of years	7
Average annual ERRs	369,232

1.12 Description of the Project Activity

The project combines solar and wind energy generation feeding the battery energy storage system (BESS) technologies to offer dispatchable and reliable power to the national electrical grid. The project is a hybrid technology project and operates as a virtual power plant, combining generation from two sites which are

over 700 km apart (900 km by road), controlled by a sophisticated virtual power plant (VPP) controller. The solar site, located at Avondale in the Northern Cape, is a 115 MW solar PV facility (utilizing thin film mono- and poly crystalline modules) feeding into a 30 MW BESS using Sungrow technology. The wind site, located at Dassiesridge in the Eastern Cape, is a 63 MW wind facility generating electricity from an array of 4.5 MW Vestas wind turbines, feeding into 45 MW of battery storage using Sungrow technology. The BESS components are located on each site prior to the grid feed substations, supplying dispatchable power to the national grid, as per the PPA. Local HPPC (Hybrid Power Plant Controller) is installed on each power plant which actively controls the assets on each site in real time. The Avondale feeds into Eskom's Avondale switching station and the Dassiesridge site feeds into Eskom's Gringley switching station. A schematic of the project is shown below.

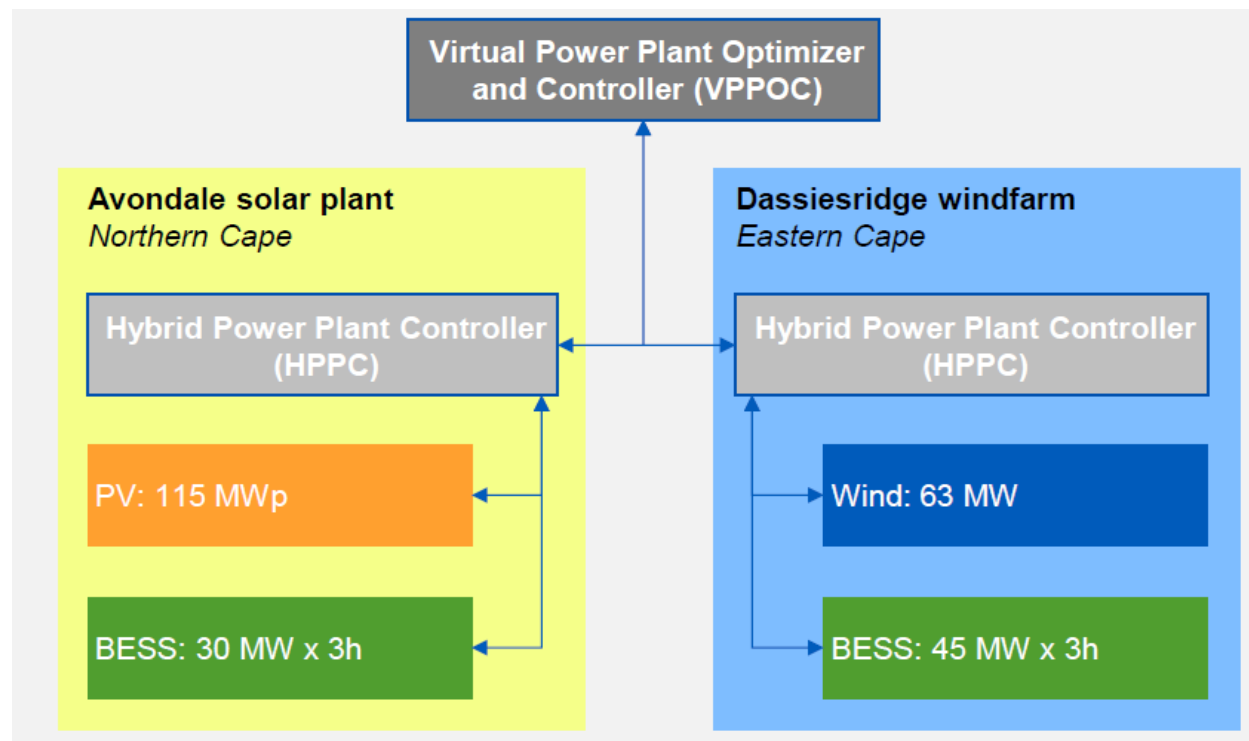


Figure 1 : Umoiyilanga plant schematic

The VPPOC uses an InAccess (Power Factors) controller that controls the inputs from the power generation to load the BESS and controls the outputs of the BESS into the national grid. Each have a local control room with dedicated personnel and are controlled centrally via the VPPOC.

The unique combination of wind and solar resources with batteries enables Umoiyilanga to provide 75 MW energy on demand from 05:00 to 21:30 daily, as per the requirements of the 20-year PPA [2], and demonstrates that renewable energy can provide reliable, dispatchable power at a competitive price.

The hybrid plant has been designed to minimize visual and noise impacts, as well as to operate safely and assuring a high level of reliability, with low water consumption and only limited maintenance and

repair with an equipment lifetime of 20-30 years, even though the power purchase agreement (PPA) duration is 20 years. Planned maintenance is scheduled on a systematic basis in accordance with OEM and good preventative and reliability maintenance practices are to be implemented for planned and unscheduled downtime.

The PV, Wind and BESS are operated remotely to optimise availability during the dispatchable period, getting most of its energy from solar during the day, and wind during the night. Excess energy is used to charge the batteries. Over the term of the PPA, Umoyilanga guarantees 84% availability over the dispatchable periods based on solar and wind resources assessment. The plant layout is shown in Figure 1 below.

Umoyilanga - Hybrid Renewable Project

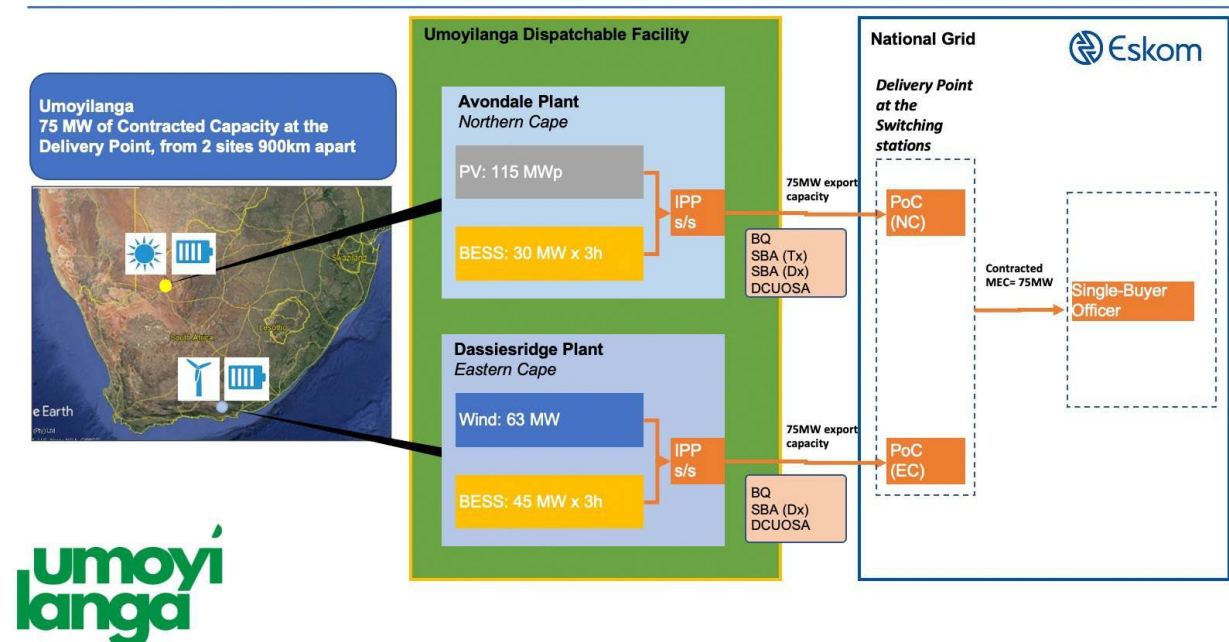


Figure 2: Umoyilanga Plant Layout

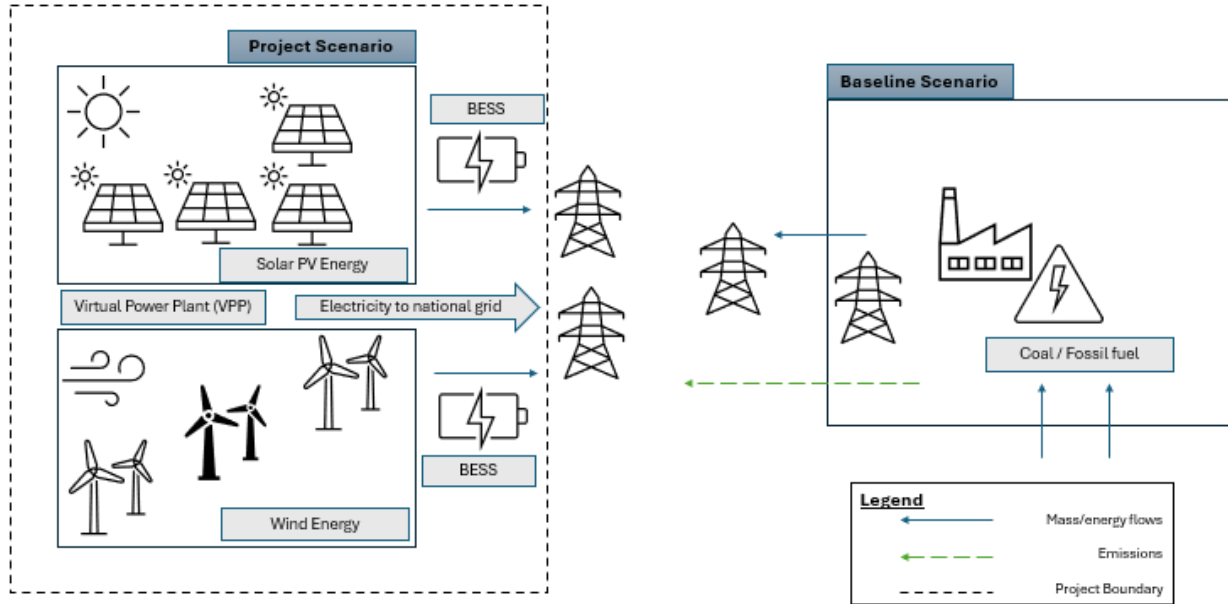


Figure 3: Schematic of project boundary

Grid Connection Scheme

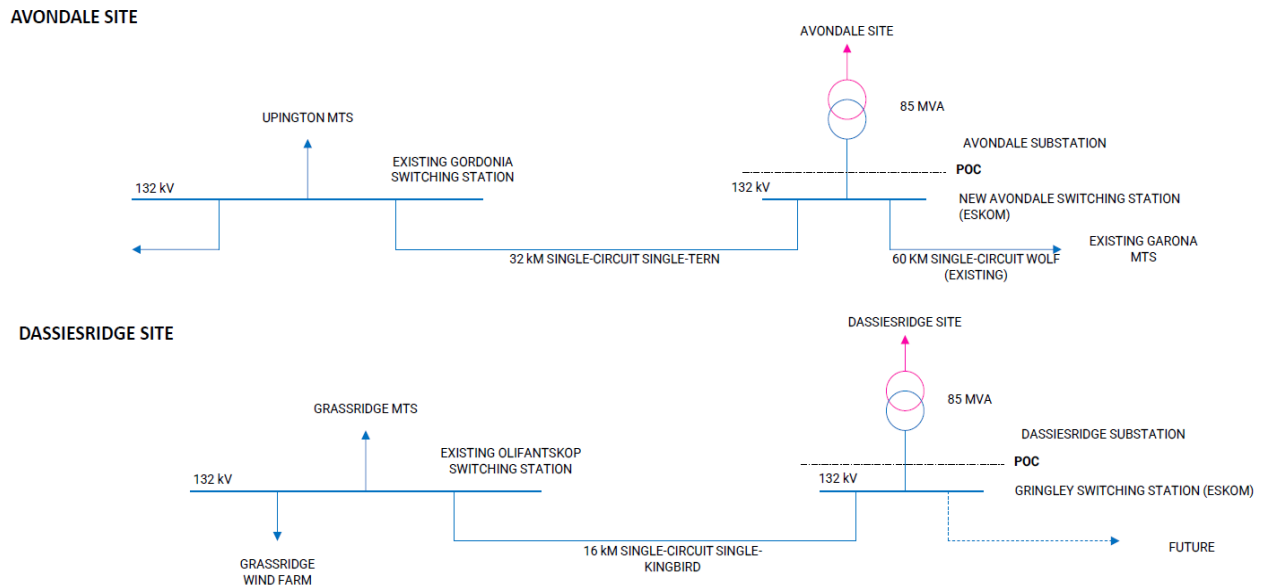


Figure 4: Grid Connection Scheme for both Sites

The main technical details of Umoyilanga Hybrid Renewable Projects are shown in the tables below.

Table 3: Proposed technologies for PV project activity

Avondale Solar PV Plant (Northern Cape)			
Technology	Type	Manufacturer (proposed)	Lifespan
Solar PV System			
PV Module	Thin-film PV modules or Mono/polycrystalline PV modules,	Seraphim (35%) Longi (65%)	20-30-years
Inverter	13 x 6,6 MVA 1 x 4,4 MVA	Sungrow	20-30-years
Transformer	MV/HV 132 kV	ACTOM	20-30-years
Trackers	1V tracker	Game Change	20-30-years
BESS /VPPOC			
VPPOC & Hybrid Power Plant Controllers	SCADA	InAccess (PowerFactors)	

Table 4: Proposed technologies for Wind project activity

Dassiesridge Wind Plant (Eastern Cape)			
Technology	Type	Manufacturer (proposed)	Life span
Wind System			
Wind turbines	V150 (4.5MW) HH120mm	Vestas	20-30-years
BESS	135 MWh	Sungrow	20-30-years
Switchboard	33 kV MV	CEDC	20-year
Transformer	MV/HV 132 kV	CEDC, Power Construction and ADENCO Construction	20-year
Virtual Power Plant Optimiser & Controller (VPPOC)			
VPPOC & Hybrid Power Plant Controllers	SCADA	InAccess (PowerFactors)	

The project planned implementation schedule is shown in Table 5 below.

Table 5: Project Implementation Schedule

NO.	Activity	Expected Finish Date
1	PV Plant	Feb/25
2	PV Substation and Switchgear	Jan/25
3	PV BESS	Feb/25
4	PV BESS Grid Compliance	Apr/25
5	Wind Plant	Apr/25
6	Wind Substation and Switchgear	Nov/24
7	Wind BESS	Jan/25
8	Wind BESS Grid Compliance	Apr/25
9	VPPOC	May/25

1.13 Project Location

The Umoyilanga project is located at two unique locations within South Africa that are connected via a Virtual Power Plant (VPP). A KML file has been provided in the VERRA registry. The locations are highlighted below.

- The Avondale site lies on Portion 1 of Farm Avondale 410, Gordonia RD within the Dawid Kruiper Local Municipality, approximately 30 km east of Upington City adjacent to the N14 national highway.

GPS geodetic co-ordinates for the site are: S 28° 21'59.93" E 021° 34'20.73"

- The Dassiesridge site lies within the Nelson Mandela Bay Municipality (NMBM), approximately 18 km northeast of Uitenhage (Kariega) town, off the R75 regional road.

GPS geodetic co-ordinates for the site are: S 33° 35'59.66" E 025° 31'14.51"

1.14 Conditions Prior to Project Initiation

The electricity generation conditions in South Africa, prior to including this Hybrid RE-BESS plant into the national grid, are described under Section 3.4 (Baseline Scenario).

Although this is not an AFOLU project, the environmental conditions are discussed as this is a Greenfield project and is established on areas previously zoned as agricultural land. The primary purpose of the project is to supply renewable and reliable electricity to South African consumers, and as such, it is not implemented with the aim of generating GHG emissions for the purpose of their subsequent reduction, removal, or destruction. Since the project comprises of a solar plant and a wind plant with associated BESS in two separate locations, the environmental conditions are discussed for both locations. Separate EIAs were conducted for each site and their associated infrastructure.

Ecosystem Type and Vegetation

The Avondale site, located within the Savannah Biome, the dominant vegetation type is a lower layer of grassland, along with woody plants spread out across the site. Notable plant species include the *Vachellia erioloba* (Camel thorn) and the *Aloe claviflora* (Kraal aloe) that trigger the requirement of a permit to removal of these protected species prior to construction. Due to the dominant vegetation being grasslands, and as the prevailing weather conditions are hot and dry, the area is prone to natural fires, although infrequent and of low intensity, which maintain the dominant grassland structure of the ecosystem [4].

The Dassiesridge site is located in the Albany Thicket Biome which is a species-rich, evergreen scrubland, home to a variety of fauna such as springbok, zebra and wildebeest. The project site traverses two vegetation types, namely the Grassridge Bontveld and the Sundays Valley Thicket which are endemic to (i.e. occur exclusively within) the Eastern Cape province). These vegetation types both provide a combination of areas dominated by well-developed woody tree species, and shrubland dominated by Fynbos. Notable species of high sensitivity include the Eastern Leopard Toad, endemic to the Eastern Cape, and avifauna such as *Aves-circus maurus* (the Black Harrier) [5].

Current and Historical Land-use

The current land-use in the Avondale project site is grazing by game and agricultural activities, in the surrounding areas close to the Orange River. The surrounding land uses also include small-scale mining.

The western half of the Dassiesridge project site is currently used for game farming, whilst the eastern half is the de-proclaimed Grassridge Private Nature Reserve, adjacent to the operational Grassridge Wind Energy Facility.

Climate

The climate for the Avondale Solar Park site can be described as warm-temperate, with annual temperatures ranging from -4 °C to 41.5 °C. The region experiences very dry winters, with the average annual precipitation varying between 120 mm and 260 mm.

The climate for the Dassiesridge site is classified as hot semi-arid with an annual average temperature of 18.2 °C and an average annual rainfall of 427 mm.

Topography

The Avondale Solar Park site is located on flat terrain, suitable for the development of a solar power plant, at an altitude of around 1,000 meters above sea level. This location benefits from high levels of solar irradiance, making it ideal for solar energy production.

The Dassiesridge site topography is characterized by moderate undulating hills, at an altitude ranging from 180 meters to 300 meters above sea level. The wind farm is located near the coastline, so benefits from strong and consistent winds.

Hydrology

The most prominent hydrological feature in close proximity to the Avondale project site is the Orange River, which runs moderately close to the southern boundary of the project site. The ridge located south of the site is a terrace that defines the floodplain of the Orange River, and drainage occurs as sheet wash towards the low-lying area, along the eastern boundary of the study area and through a drainage towards the Orange River.

A number of non-perennial rivers traverse the Dassiesridge project site, forming a significant drainage network in the region, hence the site is classified as a Strategic Water Source Area. The western half of the site occurs within 500 m of 5 wetlands, thus triggering the requirement for a Water Use Authorization to establish the infrastructure.

Soils

Based on the soils classification study at the Avondale project site, the soils are described to be weakly structured and sandy, forming a mosaic of Glenrosa soils and very shallow rocky soils. Due to their shallow nature, low pH and climatic conditions, the soils have low agricultural potential, thus making the area unsuitable for crop cultivation under arable conditions.

The soil types at the Dassiesridge site are Eutric Regosols that are typically young soils that are low in organic matter and have a low water-holding capacity, thus making them sensitive to drought and are overall unfavourable for agricultural purposes.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

Compliance with provincial, national and international legislation was considered during the planning and implementation of this project. The following legislative aspects were considered:

Table 6: Relevant Legislation

Constitution of the Republic of South Africa (Act 108 of 1996)	The Bill of Rights, in the Constitution of South Africa (No. 108 of 1996), gives everyone the right to a non-threatening environment and requires that reasonable measures are
---	--

	applied to protect the environment. This protection encompasses preventing pollution and promoting conservation and environmentally sustainable development. These principles are embraced and given further expression in NEMA.
National Development Plan, Agenda 2030	The NDP envisages that, by 2030, South Africa will have an energy sector that provides reliable and efficient energy service at competitive rates, that is socially equitable through expanded access to energy at affordable tariffs and that is environmentally sustainable through reduced pollution.
Environmental Conservation Act (No. 73 of 1989)	The ECA provides for the effective protection and controlled utilisation of the environment and for matters incidental thereto.
National Environmental Management Act, (No. 107 of 1998)	NEMA requires adherence to the principles of Integrated Environmental Management (IEM) in order to ensure sustainable development, which, in turn, aims to ensure that environmental consequences of development proposals be understood and adequately considered during all stages of the project cycle and that negative aspects be resolved or mitigated, and positive aspects enhanced.
National Water Act (No. 36 of 1998)	Provides the legal framework for the effective and sustainable management of our water resources.
Conservation of Agricultural Resources Act (No. 43 of 1983)	CARA provides for control over the utilization of the natural agricultural resources in order to promote the conservation of the soil, water sources and vegetation and the combating of weeds and invader plants; and for matters connected therewith.
Occupational Health and Safety Act (No. 85 of 1993)	OHSA requires the employer to bring about and maintain, as far as reasonably practicable, a work environment that is safe and without risk to the health of the workers.

	This means that the employer must ensure that the workplace is free of hazardous substances, such as benzene, chlorine and micro-organisms, articles, equipment, processes, etc. that may cause injury, damage or disease. Where this is not possible, the employer must inform workers of these dangers, how they may be prevented, and how to work safely, and provide other protective measures for a safe workplace.
The Basic Conditions of Employment Act (No. 75 of 1997)	The BCEA intends to give effect to the right to fair labour practices referred to in section 23(1) of the Constitution by establishing and making provision for the regulation of basic conditions of employment; and thereby to comply with the obligations of the Republic as a member state of the International Labour Organisation; and to provide for matters connected therewith.
Electricity Regulation Act (Act 4 of 2006) and the Electricity Regulation on New Generation Capacity R399 of 4 May 2011, as amended by R1366 of 04 November 2016	The ERA establishes a national regulatory framework for the electricity supply industry. It designates the National Energy Regulator of South Africa (NERSA) as the custodian and enforcer of this framework. The act covers licensing, registration and regulation of electricity generation, transmission, distribution, trading, import and export. The Regulations provide for the procurement of new generation capacity derived by organs of state, from renewable energy sources and co-generation; base load, mid-merit load and peak load new generation capacity; and cross border projects.
Integrated Resource Plan (IRP2019), and the draft IRP 2023	The IRP is the electricity infrastructure development plan, of the Department of Energy, based on the least-cost electricity supply and demand balance, taking into account security of supply and the

	environment (minimize negative emissions and water usage).
Carbon Tax Act 15 of 2019 and supporting regulations	Provide for VCUs to be used to offset carbon tax liability.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

Is the project active under the other program?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes

☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes

☒ No

1.18 Sustainable Development Contributions

1.18.1 Sustainable Development Contributions Activity Description

The sustainability objective of the project lies in its capacity to reduce greenhouse gas emissions. In alignment over the seven-year crediting period, renewable twice, the project anticipates an average annual emission reduction of 369,232 tons of CO₂ equivalent, totalling 2,584,626 tonnes of CO₂ equivalent over the first seven year crediting period and 7,384,644 tonnes of CO₂ equivalent over the project lifetime. This substantial reduction aligns with Sustainable Development Goal (SDG) 13 (Climate Action), specifically addressing the urgent need to combat climate change and its impacts. Monitoring will be undertaken as per the plan in this Project Description, which involves the monitoring of generated renewable energy in MWh/year. This aligns with South African SDG 7.2 which aims to increase the renewable energy within the country. The project displaces approximately 380,000 – 400,000 MWh per annum of emission intensive grid electricity through the generation of low emission renewable electricity.

Furthermore, aligned with SDG 8.5, the project promotes sustained, inclusive and sustainable economic growth, full and productive employment and decent work. This will be achieved through just less than 890 job opportunities during the construction phase and 70 permanent jobs during the ongoing operation of the project into the future. Given the high unemployment rate in the Avondale and Dassiesridge local communities, SDG 8.5 is a priority. These employment opportunities provided by the project will help job seekers gain experience and transferable skills, enhancing their prospects for future employment.

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	7.2	7,2,1A1 (South African SDG Indicators) – Amount of renewable energy at annual operating capacity	Implemented activities will increase the amount of renewable energy in South Africa	Displaces approximately 384,617 MWh per annum of emission intensive grid electricity through the generation of low emission renewable electricity.	Displaces approximately 7,384 GWh of emission intensive grid electricity through the generation of low emission renewable electricity.
2)	13.0	Tonnes of greenhouse gas emissions avoided or removed	The total GHG emissions of the country will be reduced by the ERs and REMs that will no longer be contributing to the country's total GHG emissions. It is expected that the number of GHG emissions will be reduced annually as the project will be verified, confirming that these are no longer emitted to the atmosphere.	Displaces approximately 369,232 tCO ₂ e per annum of emission intensive grid electricity through the generation of low emission renewable electricity.	Displaces approximately 7,384,644 tCO ₂ e of emission intensive grid electricity through the generation of low emission renewable electricity over the 20-year project lifetime.

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
3)	8.5	Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all	With total unemployment in South Africa having risen from 24.9% to 33.5% over the past decade (2012 to 2022) this SDG target remains a high priority for the country. The country has a target to, “By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value.”	During the construction phase there will be 890 job opportunities created and during Operation there will be approximately 70 permanent jobs. Furthermore, the project has committed to contributing more than 40% of the capital expenditure to local content (procurement of South African goods and services) [6].	The project planned to create 890 job opportunities during the construction phase (from domestic labour) and a further approximately 70 permanent jobs during operations [6]. The Operational jobs will be ongoing, but the construction jobs would only be during the construction phase and will phase out over a period once the plant is operational. Over the 20-year operational period, 1% of revenue will be dedicated to local communities through socio-economic initiatives [6].

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

As per the applied methodology ACM0002 (Ver. 22.0), no leakage emissions are considered in the project activity.

1.19.2 Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

1.19.3 Further Information

No further information submitted.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

The table below describes the stakeholder identification process.

Stakeholder Identification	The Environmental Impact Assessment (EIA) phase of the project took place between 2013 and 2020. This included the following aspects of the project that had separate processes: • Avondale Solar PV site • Dassiesridge Wind Energy Facility • Dassiesridge Ancillary infrastructure • Overhead power lines in the Sundays River Valley and Nelson Mandela Bay Municipalities The following measures were implemented to identify stakeholders and to request members of the public to register as interested and affected parties (I&APs): • Notices were placed in various local newspapers (Noordkaap Bulletin for the Northern Cape facility and the UD Express community newspaper for the Eastern Cape facility). • A background information document was sent to all adjacent landowners. • A background information document was distributed to stakeholders via email and placed in the local public library in Uitenhage. • Site notices were placed on the boundary of the sites. • A stakeholder and interested and affected party identification process was undertaken to create an I&AP register. A portion of the process was undertaken during the COVID pandemic, so electronic formats such as SMS, email and a dedicated website link were used.
-----------------------------------	--

	The I&AP database includes contact information, dates and details of consultations, and a record of all issues raised, and responses provided. This database will be maintained on an on-going basis. Key stakeholders include representatives from: ➤ Department of Forestry, Fisheries and the Environment (DFFE) ➤ Department of Water and Sanitation (DWS) ➤ South African Heritage Resource Agency (SAHRA) ➤ Department of Environmental Affairs: Biodiversity & Conservation ➤ Department of Energy ➤ Department of Rural Development and Land Reform ➤ Department of Science & Technology (DST) ➤ Relevant local and district municipalities ➤ Parastatals including Eskom and the CSIR ➤ Square Kilometer Array (SKA) / SA Large Telescope (SALT) ➤ Affected and surrounding landowners ➤ Environmental non-governmental organisations (Endangered Wildlife Trust, Birdlife SA) ➤ Community based organisations ➤ Residents of surrounding communities and towns
Legal or customary tenure/access rights	The project is situated in two locations where the legal tenures are private ownership, and binding lease agreements are in place with the respective landowners, in order to establish the Umoyilanga Hybrid Plant facilities on the land. Dassiesridge: EDF Renewables (the developer), on behalf of Dassiesridge Wind Power (Pty) Ltd, has engaged with the landowner and received written consent to undertake the proposed activities on the respective property. Avondale: EDF Renewables (the developer), on behalf of the Avondale Solar Wind Power site, has engaged with the landowner

	and received written consent to undertake the proposed activities on the respective property.
Stakeholder diversity and changes over time	Key stakeholders include government agencies, neighbouring landowners and residents, local communities, businesses and environmental NGOs. The Dassiesridge site straddles the Nelson Mandela Bay Metropolitan Municipality (NMBM) and Sundays River Valley Local Municipality (SRVLM). Social challenges within these municipalities include unemployment and poverty. According to the NMBM IDP, job creation and sustainable livelihood form part of the central development needs of this area [5]. The agricultural, conservation, tourism and manufacturing sectors are the key role players in the economies surrounding the project area. As such, these sectors play an important role in the social, cultural and economic fabric of the surrounding communities. The Avondale site lies within the Dawid Kruiper local municipality, where the main economic activities include agriculture, business services, game farming, tourism and hospitality [7]. Challenges experienced by residents of surrounding communities included work shortage and poverty. Approximately 45 – 50% of the surrounding communities live in poverty [8]. Changes in the make-up of the group are anticipated to be minimal throughout the duration of the project.
Expected changes in well-being	No significant negative impacts on the well-being of stakeholders are anticipated to occur because of this project. However, there are many positive impacts. Over the 20-year operational period, 1% of revenue will be dedicated to local communities through socio-economic initiatives. Furthermore, the project has committed to contributing more than 40% of the capital expenditure to local content (procurement of South African goods and services) [6]. This will result in: • Increased production / business sales (SMMes) • Increase in local Gross Domestic Product (GDP) • Increase in employment opportunities

	• Increase in skills development • Increase in household income • Improved housing and basic service provision.
Location of stakeholders	No stakeholders reside in close proximity to the project. The closest local communities that may be influenced, but are not in immediate proximity, are: • Avondale Solar ○ Karos (southern bank of Orange River) ○ Leerkrans (southern bank of Orange River) ○ Upington (limited) • Dassiesridge Wind: ○ Motherwell ○ Kariëga/Uitenhage ○ Addo
Location of resources	The Umoyilanga Hybrid facility is located on two facilities. 1. Avondale1 Solar Park - on Portion No. 1 of the Farm Avondale No. 410, Gordonia Registration Division, Dawid Kruiper Local Municipality, ZF Mgcawu District Municipality, Northern Cape Province located 34 km east of city of Upington, South Africa. 2. Dassiesridge Wind Energy Farm - is located on the following farm portions: a. Farm 185 (Elands Hoorn), Portion 11; b. Farm 187 (Grassridge); c. Farm 188 (Gringley); d. Farm 189 (Blauw Baatjes Vley), Portion 5; and e. Farm 189 (Blauw Baatjes Vley), Remaining Extent (RE). These properties are situated within the Nelson Mandela Bay Metropolitan Municipality and the Sundays River Valley

	Local Municipality, both of which are located within the Cacadu District Municipality, Eastern Cape Province. The landowners have granted Umoyilanga the rights to lease the land for the generation of electricity for the next 20 years.
--	--

2.1.2 Stakeholder Consultation and Ongoing Communication

The table below describes the process for and the outcomes from the stakeholder consultation conducted prior to project initiation.

<i>Date of stakeholder consultation</i>	11 October 2024
Stakeholder engagement process	A stakeholder engagement process formed an integral part of the scoping and Environmental Impact Assessment (EIA) phase of the project, starting in 2013 and continuing until September 2020. A database of interested and affected parties (I&APs) has been established and is kept up to date [5]. This stakeholder identification process is described in section 2.1.1 above. Public (in person) meetings could not be held at the time due to the COVID 19 pandemic restrictions, but appropriate electronic channels and mechanisms were used. Once COVID-19 restrictions were eased, multiple rounds of consultation meetings were held to share information about the project and to give stakeholders a platform to provide input to the project planning process. The meetings were held at the Tamboville Multipurpose Centre in Uitenhage on the 19 June 2024, and at the Bersheba Community Hall in Kirkwood on the 9 December 2024. The communication was presented in a culturally appropriate and inclusive manner, and inclusive language was used. For the carbon project, additional stakeholder consultation was carried out specifically in relation to the VCS requirements; building on the existing stakeholder communication process and activities. These stakeholder engagements were led by an independent external consulting company, to facilitate communication between the communities, the project developers and Anthesis Group. All stakeholders in the registered I&AP database were contacted directly via phone calls, SMS and/or emails, and printed notices were put up at the project sites.

	Advertisements were also placed in local newspapers prior to project listing inviting stakeholder and public comment.,
Consultation outcome	All issues raised during the public consultation process have been included in the issues and responses report which forms part of the EIA [4], [5] . All issues reported have been resolved and a record of all communication has been kept. The main issues raised during consultation related to the poor quality of the roads that affected the taxis and community drivers, as well as lack of job opportunities for female general workers. These issues were resolved through consultative meetings with the project proponents, where it was explained that the laborious nature of the construction phase required more male general workers, and office-based work would be made available for female workers. The construction company also committed to fixing the roads periodically when weather conditions were safe enough to work on the roads. In summary, no project or environmental related concerns were raised by registered I&APs nor have any objections been received in this regard. Additionally, the project has been received very well by all relevant authorities and no objections or opposition has been received. To date, no issues or comments have been received in relation to the VCS validation and verification process.
Ongoing communication	Ongoing communication is via direct electronic communication to stakeholders, recorded in the registered I&AP database, as well as public notices placed in accessible locations in the surrounding communities. In addition, community liaison officers have been appointed to facilitate communication with local communities on an ongoing basis. All grievances received are recorded and addressed as far as practically possible.
Stakeholder input	All comments received during the EIA and stakeholder consultation have been carefully considered and addressed in the EIA and project implementation. No issues or comments received were directly related to the VCS validation and verification process, therefore no updates to the project design have been considered necessary.

Stakeholder comments will continue to be monitored to consider whether further updates to the project are required.



Plate 5.1: Proof of Site Notice board placed at 33°35'58.55"S, 25°31'25.84"E.

Figure5: Example of public notice placed on site

from Kariega to the world for 32 years, and grounds and sectors.

Quintin Levey (Exporters EC Chairman).

PHOTO: SUPPLIED

Dis heerlike Nartjiefees dié Saterdag!

DIE lekkerste en beste gesinsfees in die Oos-Kaap – die Nartjiefees – vind weer Saterdag, 19 Oktober op Loerie plaas!

Maak seker jy is daar vir al die lekker musiek deur Fatman, Ruan van der Kuil, Dillon Lerm en Chris en Elmarie Brits.

Vir die honger maag is daar smullekker kos, gebak, poeding, pannekoek, tannie Joan se beroemde tuisgemaakte konfyt, plaasvares groente om saam huis toe te neem en natuurlik heerlike nartjies!

Kuier in die saal se koffiewinkel met ou en nuwe vriende terwyl die kleinspan die kindervermaak geniet. Staan nader vir die nartjie-eet kompetisie of snuffel rond by die kunsmarkstalletjies.

Die Nartjiefeessterrein beskik oor veilige parkeerplek, en voorsiening vir gestremde persone is aan die onderkant van die groentestalletjie gemaak. Die hekke open om 08:00 en die fees behoort teen 15:30 klaar te wees. Toegangsfioe is R120 vir volwassenes en R80 vir leerdere. Voorskoolse kinders – vergesol van hul ouers – kom gratis fees toe. Nartjies is by die hek en op Quicket beskikbaar.

Kaartfasiliteite is by die fees beskikbaar. Bring opvoorstee saam om na die kunstenaars te luister. Geom piknikmandjies of gazebo's word op die terrein toegelaat nie.

Verdere inligting is op Facebook en www.nartjiefees.co.za beskikbaar. Ander navrae kan by 083 383 4932 verkry word.

Umoyilanga Project

Umoyilanga Hybrid Renewables Project to boost SA's Clean Power Supply in 2025

The Umoyilanga hybrid renewables battery energy storage project is set to begin operations next year, integrating solar, wind, and battery storage technologies to deliver 75MW of reliable, dispatchable, clean power to South Africa's national electricity grid.

Functioning as a virtual power plant, the project will harness energy from two sites – Avondale in the Northern Cape and Dassenburg in the Eastern Cape. It is expected to create approximately 150 full-time jobs during the construction phase.

Owned by EDF Renewables and Perpetua Holdings, Umoyilanga will produce electricity under a 20-year power purchase agreement notified in August 2023. This initiative is part of the South African government's Risk Mitigation Independent Power Producer Procurement Programme. It will reduce greenhouse gas emissions by replacing fossil fuel derived electricity with clean renewable energy and, as such, deliver carbon credits under the Verified Carbon Standard (VCS). These credits are developed by Arthesis and can be used by companies to offset their carbon tax obligations. Independent auditors will be validating the project to ensure it meets the strict international requirements of the VCS. Independent auditors will be validating the project to ensure it meets the strict international requirements of VERRA.

For more information about this project, or to raise any questions or concerns, please email projectsza@anthesisgroup.com

About Arthesis: Arthesis empowers clients to achieve sustainable performance as the world's leading, purpose-driven, digitally enabled, science-based actiivator. In South Africa, our core offerings include Carbon Project Development, Carbon Tax Advisory, and Carbon Offsetting, all designed to help organizations navigate their environmental responsibilities and drive impactful change.



Loerie Nartjiefees 2024

19 OKTOBER, NG KERK LOERIE

09:00 DILLON LERM,
10:30 CHRIS BRITS
11:30 RUAN VAN DER KUIL,
12:30 NARTJIE-EET
13:15 CHRIS & ELMARIE BRITS,
14:00 FATMAN

HEELTYD FEESTYD, FAMILIE VERMAAK, LEKKER KOS!

Quicket HEKKE OPEN 8AM
KAARTJIES BESKIKBAAR **R120** **R80**
VIR MEER INLIGTING KONTAK 083 383 4932
WWW.NARTJIEFEES.CO.ZA

Bring opvoorstee, saam, om die kunstenaars te luister, geom piknikmandjies of gazebo's word op die terrein toegelaat.

Figure6: Example of Newspaper advert (UD Express - October 2024)

Umoyilanga Hybrid Renewable Energy Project - stakeholder consent

Projects ZA <projectsza@anthesisgroup.com>
To: LUAFhelpdesk@daird.gov.za

Dear Stakeholder

We are contacting you as someone who has an interest in or may be impacted by the Umoyilanga Hybrid Renewable Energy Project (Umoyilanga). As we move forward with developing carbon offsets from the project's emission reductions, we are starting a new phase of stakeholder consultation.

Please would you review the information provided below and provide consent to participate in the stakeholder consultation via the link below. You may also provide comments below too if you would like to.

ACTION REQUIRED - Consent

Please visit this link to inform us if you are willing to provide consent in the stakeholder consultation process: <https://forms.office.com/r/2P0gUdU>

Umoyilanga Stakeholder Information

Project Design and Implementation

The project combines solar, wind and battery storage technologies to offer dispatchable and reliable power to the national Eskom electricity grid. The project will operate as a virtual power plant, combining generation from two sites which are over 700 km apart, namely Avondale, near Upington in the Northern Cape, with 115 MW of solar PV and 30 MW of battery storage, and Dassenburg, near George in the Eastern Cape, with 65 MW of wind and 40 MW of battery storage.

The unique combination of wind and solar resources with batteries, enables Umoyilanga to provide 75 MW of demand from 05:00 to 21:30 into South Africa's national power grid, and demonstrates that renewable energy can provide reliable, dispatchable power at a competitive price, not just during daylight hours or when the wind is blowing.

The project will be replacing coal based electricity with renewable energy and hence will result in a reduction in Greenhouse Gas (GHG) emissions. These GHG emission reductions will be quantified and developed into Verified Carbon Units (VCUs) under the Verified Carbon Standard (VCS) program, administered by Verra. These are also more commonly known as carbon credits. It is envisaged that the project will come online during the second quarter of 2025.

Risks, Costs and benefits the project may bring to stakeholders

A comprehensive analysis of the risks associated with the project was conducted as part of the approved Environmental Impact Assessments.

No high risks were identified, potential moderate risks to the neighbouring communities include:

- Impact on flora and fauna, especially during construction
- Localised noise, especially during construction
- Potential for limited erosion, especially during construction
- General pollution & inconvenience associated with the construction site

Umoyilanga has taken measures to mitigate all these risks to a low level, and to rehabilitate the site as far as possible as stipulated in its approved Environmental Management Programme, and therefore no significant negative impacts to the surrounding communities are envisaged. No significant risks were identified for once construction is completed and the project is in operation.

Furthermore, the project will have a range of positive effects on surrounding communities including job creation, skills development, improving infrastructure, and adding the transition to a low carbon economy.

As a developer and employee, the Umoyilanga project respects the rights of the stakeholders and employees and upholds the international bill of human rights. If you feel that the project is infringing on any human rights, please notify us via our contact page provided at the end of the document.

Laws and regulations covering workers' rights in South Africa

Workers' rights in South Africa are dictated by the Basic Conditions of Employment Act (BCEA, Act 75 of 1997), as well as the Occupational Health and Safety Act (Act 85 of 1993) which ensure that all employees are treated fairly and in line with the national legislation as a minimum. If you feel that the project is violating these laws, please notify us by replying to this email or visit our comments page on our website: <https://www.anthesisgroup.com/za/project-comment/>

Envisaged impact on property rights

This project is constructed on privately-owned land, which is leased from the landowner. The legally binding lease agreement and site deed held by the owner, are proof that the project has not encroached on land outside this boundary. No people have been relocated without consent and no physical or economical displacement has occurred as a result of the project, no impact on any protected area has taken place. Any risks to or impacts on adjacent landowners or inhabitants were assessed and mitigated as part of the EIA process.

The process of generating Voluntary Carbon Standard (VCS) validation and verification and the validation/verification body's site visit

The process of generating VCUs through the VCS system involves the following steps:

- Project Development & Validation:** The project participant appoints an independent auditor (validation and verification body - VVB) to audit the project against the international requirements. The project now enters validation, including a 30-day public comment period. A project description document is listed on Verra's website as "under validation".
- Project Registration:** Once the project has successfully completed validation and met all the requirements, it can request registration, and status on Verra's website is updated to "registered".
- Project Monitoring:** After the project has been operational for some time, the project can progress to project verification. This step requires a project monitoring report and also requires another independent audit. It is during this phase that the project may only request VCUs for the first time.
- Initial VCU Issuance:** Once the project has successfully completed verification it can request initial issuance of VCUs.
- Periodic VCU Issuance:** VCUs are issued every monitoring round (usually annual intervals of a crediting period). The crediting period has been selected as three rounds of seven years (so a total of up to 21 years). After each crediting period, the project undergoes validation again.
- VCU Retirement:** The Verra Registry displays the status of every VCU issued under the VCS program. VCUs may have a status of active, retired or cancelled.
- Project Maintenance:** The project participant ensures that all project details are kept up to date and notifies Verra of any changes in project details.

ACTION REQUIRED - Comment

Figure 74: Example of an email sent to stakeholders

2.1.3 Free Prior and Informed Consent

Obtaining consent	The landowners have consented and granted rights to the project to install the associated energy and storage infrastructure on the properties and lease agreements are in place. No IPs (Indigenous Peoples), LCs (Local Communities) or customary rights holders are impacted by the Umoyilanga Hybrid project activities.
Outcome of FPIC	The outcome of the process is that the Umoyilanga Hybrid facility will be established on private land for the next 20 years. No other land resources or communities have been negatively impacted.

2.1.4 Grievance Redress Procedure

Development process	Grievances can be received via three channels: • The grievance portal on the Anthesis website • Community Liaison Officers (CLOs), permanently employed (or contracted) within the operating PP team, are stationed in the surrounding communities and are tasked with keeping communication up to date with Umoyilanga, for the lifetime of the project; and • The project proponents have placed grievance boxes in the communities where stakeholders can submit their comments or concerns about the project, and these are maintained by the CLOs. Any grievances received by Anthesis will be handled by Anthesis in consultation with the Umoyilanga project team, according to the procedure described below. Those received by the Umoyilanga team will be handled on site and be communicated with Anthesis periodically.
Grievance redress procedure	Any grievances, complaints or conflicts from stakeholders would be captured via a dedicated feedback form on the Anthesis website, or from the other channels mentioned above. A grievance register is maintained for site management. The Grievance Redress process consists of:

	- Receiving the issue from a stakeholder - Assigning it to the relevant manager/team - Responding to the stakeholder (which may include telephonic, written or face-to-face communication) in order to understand and resolve the issue in good faith - Recording the outcome of the interaction. - Grievances that cannot be resolved amicably in this manner will be referred to mediation by a neutral third party and failing resolution shall be raised to either arbitration or courts of law in South Africa. Stakeholders have an ongoing opportunity to log into a dedicated feedback form on the Anthesis website in order to leave comments and engage with the project team. This website will be part of the ongoing and continuous engagement between the project and stakeholders. Comments received will be logged, reviewed and taken into account in ongoing project operation, management reporting and future verifications.
--	---

2.1.5 Public Comments

Comments received	Actions taken
No comments were received during the public commenting period (global stakeholder consultation period on Verra's website).	No changes to the project were required since no public comments were received during this period.
No issues raised or comments received from local stakeholders were directly related to the carbon project or the VCS validation and verification process.	No updates to the project design have been considered necessary,

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

The Anthesis SA team has significant experience in implementing similar project activities, engaging with communities, project management and environmental and social safeguards

aligned with the VCS requirements. Members of the project development team have as a collective, in excess of 50 years' experience in the voluntary carbon market. They have been originating, developing and monetizing carbon credits for over 2 decades. The Umoyilanga project proponent has extensive renewable energy project experience in South Africa region, including similar Renewable Power facilities at Waainek; Chaba and Wesley-Ciskei. Their parent company is a global power company with experience in projects such as Aramon Solar PV plant, France; Maverick Solar PV plant, USA; Al Dhafra Solar PV plant, UAE; Ventos de Bahai wind, Brazil and Dumat Al Jandal Wind, Kingdom of Saudi Arabia.

The integrated team is well positioned to implement and operate the facility and deliver the associated GHG emission reductions into the local market.

2.2.2 Risk Assessment

	Risks identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing	No natural and human induced Risks	Measures to prevent natural and human-induced risks were documented in the EIA report.
Risks to stakeholder participation	No risks identified	Stakeholder engagement process was conducted, and comments received from the stakeholders were properly addressed. No evidence of participation barriers to stakeholders identified. The process for stakeholder's participation was undertaken as required by the South African EIA regulations.
Working conditions	No risks identified related to working conditions	All workers and their working conditions are subject to the Basic Conditions of Employment Act of South Africa and labour conditions are covered by periodic inspections and compliance to the South African Department of Labour.
Safety of women and girls	No risks related to women and girls identified	PP has taken all measures to

		ensure safety of women and girls by implementing various policies as per South African labour and anti-discrimination laws.
Safety of minority and marginalized groups, including children	No risks identified	PP ensures safety of minority and marginalized groups, including children in the project site as per the South African labour and anti-discrimination laws.
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	During Construction phase, increased traffic volumes, land disturbance and people will lead to increased levels of air pollution and noise. During Operational phase, maintenance activities may require the use of hazardous substances (like lubricants and oils), and herbicides may be used for vegetation control.	An Environmental Management Programme (EMPr), approved by government authorities as part of the EIA process, is in place for the Construction as well as the Operational phases. The EMPr addresses training and awareness and provides mitigation measures for pollution impacts, waste management, fire risks and other environmental matters.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified ⁵	Mitigation or preventative measure(s) taken
Discrimination	No risks identified	PPs have formal human resource and labor policies as per the Basic Conditions of Employment Act of South Africa.
Sexual harassment	No risks identified	PPs have formal human resource and labor policies as per the Basic Conditions of Employment Act of South Africa.

⁵ The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

Equal pay for equal work	No risks identified	PPs have formal human resource and labor policies as per the Basic Conditions of Employment Act of South Africa.
Gender equity in labor and work	No risks identified	PPs have formal human resource and labor policies as per the Basic Conditions of Employment Act of South Africa.
Forced labor	No risks identified	PPs have formal human resource and labor policies as per the Basic Conditions of Employment Act of South Africa.
Child labor	No risks identified	PPs have formal human resource and labor policies as per the Basic Conditions of Employment Act of South Africa and the Bill of Rights entrenched within the Constitution of South Africa.
Human trafficking	No risks identified	PPs have formal human resource and labor policies as per the Basic Conditions of Employment Act of South Africa and the Bill of Rights entrenched within the Constitution of South Africa.

2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
No risks identified	PPs adhere to international and national human right laws and the Bill of Rights entrenched within the Constitution of South Africa. Risks related to human rights were assessed during the EIA process.

2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
Sites of archeological or cultural significance may exist within project boundary	An archaeological and heritage assessment, conducted by AGES as part of the EIA process, determined the project impacts on heritage resources to be of LOW significance. All the heritage sites found on the property are to be excluded from the PV plant footprint and a minimum 30 m buffer zone is to be maintained.

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
No risks identified	The project is established on private property and land tenure is secured through legally binding lease agreements. No at-risk communities were involved or affected.

2.3.5 Benefit Sharing

The project does not impact on property rights and therefore no benefit sharing agreement is required

2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems	Potential loss of endangered plant species during construction of pylons and infrastructure.	Botanical specialists identified No-Go Areas and delineated where pylons and infrastructure cannot be installed, and Ornithologists advised on bird-friendly infrastructure to be used ⁶⁶ . The

⁶⁶ <https://edf-re.co.za/successful-translocation-of-eagle-nests/>

		EMPr addresses biodiversity impacts.
Soil degradation and soil erosion	Clearance of vegetation may increase run-off and lead to top-soil erosion	EMPr includes Effective Monitoring System for erosion risks and revegetation and re-seeding of disturbed areas for erosion control and rehabilitation.
Water consumption and stress	- Pollution of hydrological features due to erosion risks during storm events. - Leakage of sanitation system into groundwater	EMPr includes storm water management infrastructure to be maintained and cleared of debris, and to ensure no leakage of wastewater into groundwater and surrounding hydrological resources.

2.4.1 Rare, Threatened, and Endangered Species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☒ Yes

☐ No

Sensitive species and habitats are listed and discussed in detail in the environmental impact assessment reports for both sites [5] [4]. These documents also propose mitigation measures for the construction and operational phases, which were considered by the relevant authorities and stakeholders prior to Environmental Authorization being granted [9],[10]. The Environmental Management Programme (EMPr) for Dassiesridge [11] and Avondale [12] provide further measures to control and mitigate environmental impacts.

Information presented in the table below shows that environmental impacts have been duly identified and adequate mitigation measures are in place to address these.

Species and habitat	Avondale solar PV and BESS facility is not located within a threatened ecosystem, as the Gordonia Duneveld has a conservation status of Least Threatened with very little transformation [4]. Protected trees occurring at Avondale, requiring a license from the government authority to remove them prior to construction, include:
---------------------	---

Vachellia erioloba (Camel thorn), and Boscia albitrunca (Shepherds tree).

Plants protected by provincial legislation, requiring a permit for removal prior to construction, include:

Aloe claviflora (Schedule 2 species);
Boscia foetida (Schedule 2 species);
Harpagophytum procumbens (Schedule 1 species).

The Avifaunal Impact Assessment for Avondale reports 19 red listed birds (six classified as vulnerable, and the rest as near threatened), and 13 red listed mammals, have historic distribution ranges in the area. No protected amphibians or reptiles occur in the area.

The Avifauna Impact Assessment for Avondale concluded that, provided the suggested mitigation measures and recommendations are adhered to, it is unlikely that the proposed development will have a long-term, significant negative impact on the local avifauna, although a monitoring plan should be implemented during the construction and operational phases of the PV plan

The Dassiesridge Wind and BESS facility is not located within a threatened ecosystem, as the site is situated approximately 3.5 kilometres south of an Endangered Ecosystem (Albany Alluvial Vegetation) [13]. Although the Grassridge Bontveld and Sundays Valley Thicket occur exclusively in the Eastern Cape Province, they are both classified as Least Concern, because they are Moderately Protected. The National Screening Tool Report for the Dassiesridge BESS (October 2020) classifies the entire development site as having:

- 1) very high terrestrial biodiversity sensitivity (as the site is in a near-natural state (classified as CBA 2⁷)
- 2) low plant species sensitivity; and
- 3) high animal species sensitivity, due mainly to the presence of *Neotis denhami* (Denham's bustard) a bird categorised as Near Threatened on the IUCN Red List (Global status) and as Vulnerable according to the Red Data Book of Birds (Regional status). Other animal species, classified as 'medium sensitivity' which are likely to occur in the area include:

→ *Circus maurus* (Black harrier), a bird categorised as Endangered on the IUCN Red List (Global status);

→ *Aneuryphymus montanus* (Yellow-winged Agile Grasshopper), an insect categorised as Vulnerable on the IUCN Red List; and

→ a reptile "Sensitive species 18"⁸.

The Ecological Impact Assessment report for Dassiesridge lists 16 plant species in the area as protected, 12 mammals (eight threatened species and four near threatened species), and 4 reptiles endemic to the Eastern Cape Province. A permit to remove protected plants was obtained from the provincial authority prior to construction.

⁷ Critical Biodiversity Areas (CBAs) are classified in provincial spatial biodiversity plans, where 'CBA 2' areas are valuable for conservation; for maintaining ecological processes, sustaining biodiversity, and supporting ecosystem services. These areas may support species or habitats that are important for the broader landscape's ecological health. Whereas CBA 1 areas are essential for conservation.

⁸ Some species of conservation concern (SCC) are sensitive to illegal harvesting, so their names are obscured, as per the best practice guideline that accompanies the National Protocol and Screening Tool. The name of the sensitive species may not appear in any specialist reports released into the public domain.

	The Avifauna Impact Assessment records 50 bird species, classified as regionally and Globally Red Listed, or endemic to the province, and which are likely to occur in the area of development. Four bird species in the province are listed as endangered, eight species are vulnerable, and eight species are listed as Near threatened. The Ornithologist expressed an opinion that the project may proceed with acceptable levels of risk to avifauna, provided that the recommendations of their report are implemented.
Areas needed for habitat connectivity	The impact studies did not identify areas needed for habitat connectivity, and biodiversity offsets were not called for by the authorities. The professional opinion expressed in both Environmental Impact Assessment reports is that no fatal flaws have been identified, to prohibit the respective environmental authorizations from being granted. Careful consideration was given to identify environmental sensitivities, and mitigation measures were recommended in various specialist studies.

	Risks identified	Mitigation or preventative measure(s) taken
Habitats for rare, threatened, and endangered species	Risks and impacts identified in the EIAs are comprehensively assessed	Management interventions and mitigation measures are proposed in the EMPr for both facilities, and include:

	and classified and include the following: Ecosystem disruptions and impacts due to vegetation clearance and earthworks in the construction phase Dust and air pollution from earthworks and vegetation clearance during the construction phase. Veld fire impacts to habitats and infrastructure Pollution to groundwater, surface water and land from erosion, poor sanitation, inadequate waste management, fuel and chemical spills. Solar and wind infrastructure impacts avifauna species.	Sensitive areas to be protected and remain undeveloped. Search and Rescue operation(s) for species of conservation concern to be relocated prior to construction Construction activities and vegetation clearance restricted to the development footprint Topsoil removed, stockpiled and used to rehabilitate disturbed areas. Control of exotic invasive plant species Vegetation control (for fire risks and alien species invasion) Bird and bat friendly infrastructure to minimize collisions and electrocutions Adequate and functioning sanitation systems on site Proper waste management systems
Areas for habitat connectivity	No risk identified	No mitigation /preventive measures proposed for habitat connectivity

2.4.2 Introduction of Species

Not applicable, as there is no introduction of new species in the construction or operation of the project activity.

The EMPr contains management interventions and mitigation measures for controlling alien invasive species.

Existing invasive species	Mitigation measures to prevent the spread or continued existence of invasive species
<i>Opuntia ficus-indica</i>	EMPr incorporates removal through biological and mechanical control.

	• Alien Invasive Control Programme will be implemented to monitor spreading of alien invasive species.
<i>Opuntia aurantiaca</i>	• EMPr incorporates removal through biological and mechanical control. • Alien Invasive Control Programme will be implemented to monitor spreading of alien invasive species.
<i>Acacia mearnsii</i>	• EMPr incorporates removal through mechanical or chemical control. • Alien Invasive Control Programme will be implemented to monitor spreading of alien invasive species.
<i>Acacia cyclops</i>	• EMPr incorporates removal through mechanical or chemical control. • Alien Invasive Control Programme will be implemented to monitor spreading of alien invasive species.

	Risks identified	Mitigation or preventative measure(s) taken
Invasive species	Invasive alien plants pose a direct threat to indigenous biological diversity, water security, ecological functioning of natural systems and the productive use of land.	. Management interventions and mitigation measures are documented in the EMPr, and include: No invasive species may be introduced Established invasive species must be controlled in accordance with an Alien Invasive Control Management Plan, which must be in place.

2.4.3 Ecosystem Conversion

This section is not applicable, seeing that this is not an ARR, ALM, WRC or ACoGS project.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	ACM0002	Grid-connected electricity generation from renewable sources	22.0
Tool	TOOL 01	Tool for the demonstration and assessment of additionality	7.0
Tool	TOOL 07	Tool to calculate the emission factor for an electricity system.	7.0
Tool	TOOL 24	Common practice	3.1
Tool	TOOL 27	Tool for the Investment analysis	14.0

3.2 Applicability of Methodology

The project activity(s) meets each of the applicability conditions of the methodology(s), tools, and modules applied by the project (where applicable), as described below:

Methodology ID	Applicability condition	Justification of compliance
ACM0002	This methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plant(s)/unit(s);	The project activity involves installation of a greenfield power plant that is connected to the national grid of South Africa. So, this meets the applicability condition (a).

Methodology ID	Applicability condition	Justification of compliance
	(d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s)/unit(s); or (f) Install a Greenfield power plant together with a grid-connected Greenfield pumped storage power plant. The greenfield power plant may be directly connected to the PSP or connected to the PSP through the grid.	
	In case the project activity involves the integration of a BESS, the methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Integrate BESS with a Greenfield power plant; (b) Integrate a BESS together with implementing a capacity addition to (an) existing solar photovoltaic¹ or wind power plant(s)/unit(s); (c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any other changes to the existing plant(s); (d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power plant(s)/unit(s); (e) Integrate a BESS together with a Greenfield power plant that is operating in coordination with a PSP. The BESS is located at site of the greenfield renewable power plant.	This project activity is a grid connected renewable energy power generation project which includes the integration of a BESS managed by a virtual power plant controller into the greenfield power plant. So, the project meets applicability condition (a)
	The methodology is applicable under the following conditions:	

Methodology ID	Applicability condition	Justification of compliance
	(a) Hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit must have started commercial operation prior to the start of a minimum historical reference period of five years. The reference period is used for the calculation of baseline emissions and defined in the baseline emission section. Furthermore, no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity; (c) In case of Greenfield project activities applicable under paragraph 7(a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g., by referring to feasibility studies or investment decision documents); (d) The BESS should be charged with electricity generated from the associated renewable energy power plant(s). Only during exigencies may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions	(a) This project is a hybrid renewable energy plant utilising solar power and wind power. (b) This project is not a capacity addition, retrofit, rehabilitation or replacement, but a greenfield plant. (c) BESS is an integral part of the project design and is a requirement of the project approvals and power purchase agreement⁹. (d) BESS will only be charged from the greenfield renewable energy (solar and wind) plants at each location. No grid power will be used to charge the BESS component.

⁹ Reference to PPA

Methodology ID	Applicability condition	Justification of compliance
	following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g., week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period. (e) In case the project activity involves PSP, the PSP shall utilize the electricity generated from the renewable energy power plant(s) that is operating in coordination with the PSP during pumping mode.	(e) Not applicable to this project as the project does not employ PSP
	In case of hydro power plants, one of the following conditions shall apply: (a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or (b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation (7), is greater than 4 W/m²; or (c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m²; or (d) The project activity is an integrated hydro power project involving multiple	Not applicable to this project as the project does not employ hydro power

Methodology ID	Applicability condition	Justification of compliance
	reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m², and all of the following conditions shall apply: (i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m²; (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m² are: a. Lower than or equal to 15 MW; and b. Less than 10 per cent of the total installed capacity of integrated hydro power project.	
	In the case of integrated hydro power projects, project participants shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or (b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the	Not applicable to this project as the project does not employ hydro power

Methodology ID	Applicability condition	Justification of compliance
	specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.	
	In the case of PSP, the project participants shall demonstrate in the VCS Project Description that the project is not using water which would have been used to generate electricity in the baseline.	Not applicable to this project as the project does not employ PSP
	The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; (b) Biomass fired power plants/units.	(a) Not applicable to this project as the project does not employ any fuel switching. (b) Not applicable to this project as the project does not employ any biomass firing
	In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance”.	Not applicable to this project as the project does not employ any retrofits, rehabilitations, replacements, or capacity additions

Methodology ID	Applicability condition	Justification of compliance
CDM TOOL 01	The use of the “Tool for the demonstration and assessment of additionality” is not mandatory for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool.	Project activity does not entail proposing a new methodology. As such TOOL 01 is mandatory and therefore was applied to prove additionality.
CDM TOOL 01	Once the additionality tool is included in an approved methodology, its application by project participants using this methodology is mandatory.	TOOL 01 is referenced in the methodology ACM0002 as the prescribed tool to prove additionality and, as such it is mandatory for this project.

Methodology ID	Applicability condition	Justification of compliance
CDM TOOL 07	This tool may be applied to estimate the Operating Margin (OM), Build Margin (BM) and/or Combined Margin (CM) when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	This project will feed electricity into the South African national grid, and as such, this is the prescribed tool to calculate the baseline emissions.
CDM TOOL 07	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case,	This project does not include any off-grid power plants. Therefore, this tool was used to calculate the emission factor for the project

	two sub-options under step 2 of the tool are available to the project participants, i.e. option II a and option IIb. If option II a is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	baseline electricity system for grid power plants only.
CDM TOOL 07	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	The project activity is located in South Africa (a non-Annex I country) and hence the tool is applicable.
CDM TOOL 07	Under this tool, the value applied to the CO ₂ emission factor of biofuels is zero.	The project activity does not involve biofuels.

Methodology ID	Applicability condition	Justification of compliance
CDM TOOL 24	This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, or baseline and monitoring methodologies that use	Project activity applies methodological tool: Tool for the demonstration and assessment of additionality and therefore the Common Practice tool is applicable to the project activity.

	the common practice test for the demonstration of additionality.	
CDM TOOL 24	In case the applied approved baseline and monitoring methodology defines approaches for the conduction of the common practice test that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	Methodology ACM0002 does not define separate approach for the conduction of common practice test, which are different from the ones on the tool.

Methodology ID	Applicability condition	Justification of compliance
CDM TOOL 27	This methodological tool is applicable to CDM project activities and programmes of activities (PoAs) that conduct an investment analysis for the demonstration of additionality and/or the identification of the baseline scenario	Project activity complies as it does use the investment analysis to demonstrate additionality. This is presented in §3.5 below.
CDM TOOL 27	In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	Project activity complies as methodology ACM0002 provides for the use of investment analysis.

3.3 Project Boundary

As per the methodology ACM002 version 22, the project boundary includes the power plants, and all power plants connected physically to the electricity system that the project power plant is connected to.

Source		Gas	Included?	Justification/Explanation
Baseline	Grid Connected Electricity Generation	CO ₂	Yes	Main emission source (Grid electricity)
		CH ₄	No	Excluded for simplification. Minor source of emissions. Conservative
		N ₂ O	No	Excluded for simplification. Minor source of emissions. Conservative
		Other	No	No other GHG emissions relevant
Project	Greenfield Solar Photovoltaic electricity generation Project Activity	CO ₂	No	No CO ₂ emissions from project activity. Minimal taken as negligible
		CH ₄	No	Excluded. Minimal taken as negligible
		N ₂ O	No	Excluded. Minimal taken as negligible
		Other	No	No other GHG emissions
	Greenfield Wind electricity generation Project Activity	CO ₂	No	No CO ₂ emissions from project activity. Minimal taken as negligible
		CH ₄	No	Excluded. Minimal taken as negligible
		N ₂ O	No	Excluded. Minimal taken as negligible
		Other	No	No other GHG emissions
	Charging of BESS using electricity from the renewable energy generation plants.	CO ₂	No	Not accounted as BESS will only be charged from the renewable energy generation plants on site therefore taken as negligible.
		CH ₄	No	Not accounted as BESS will only be charged from the renewable energy generation plants on site therefore taken as negligible.
		N ₂ O	No	Not accounted as BESS will only be charged from the renewable energy generation plants on site therefore taken as negligible.

Source	Gas	Included?	Justification/Explanation
	Other	No	Not accounted as BESS will only be charged from the renewable energy generation plants on site therefore taken as negligible.

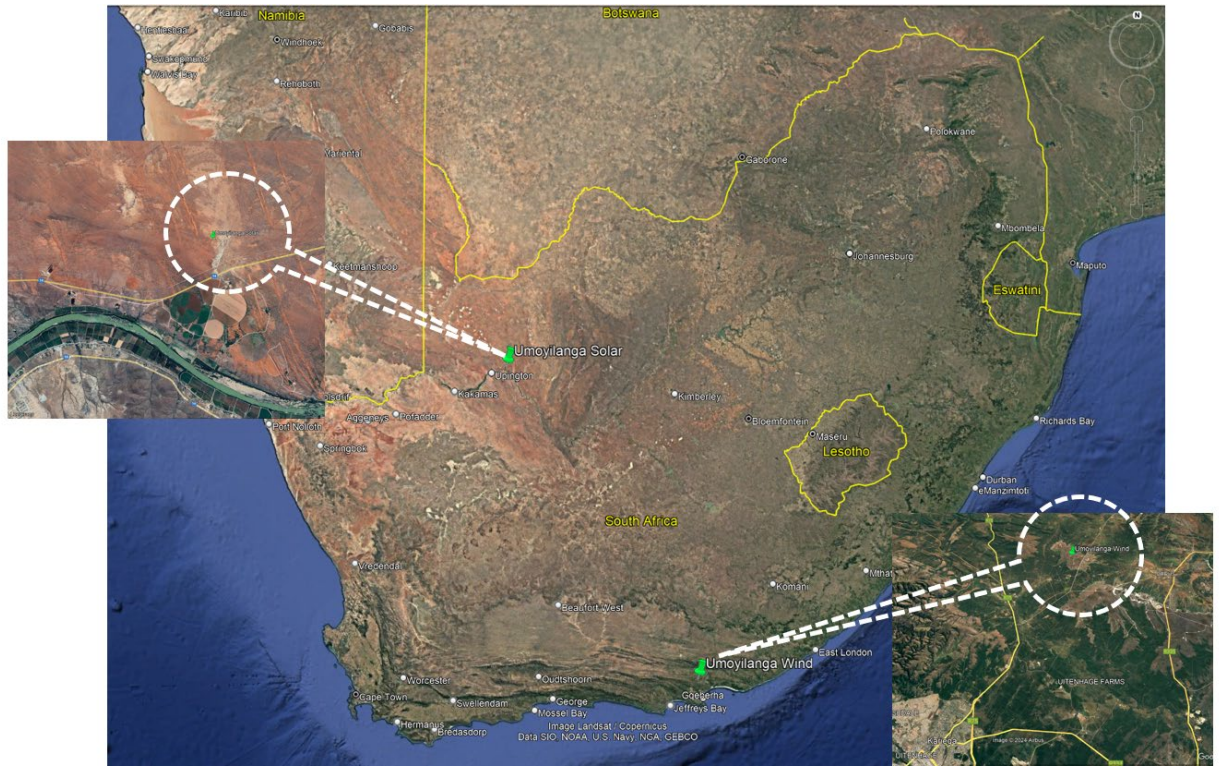


Figure 5: Physical location of the project activities

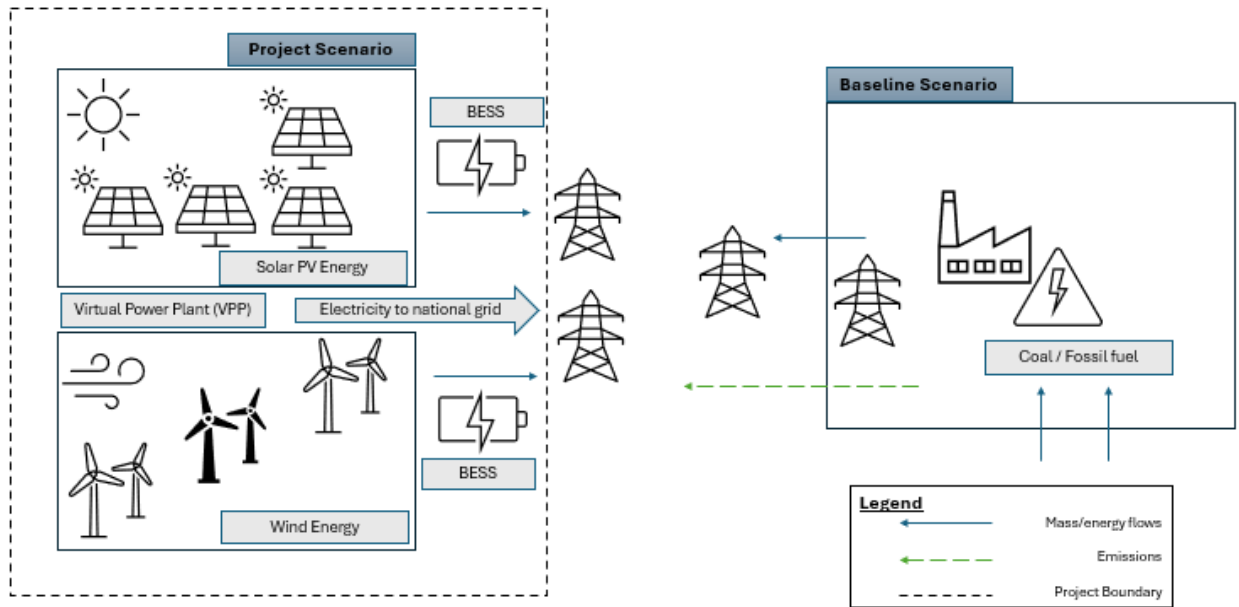


Figure 6: Project boundary, mass & energy flows of Umoyilanga Hybrid Project

3.4 Baseline Scenario

The baseline scenario for the Umoyilanga Hybrid project is defined as the continuation of current grid operations, where electricity demand is primarily met by existing grid-connected fossil fuel (predominantly coal) power plants, and the addition of new fossil fuel-based capacity/generation sources. This approach aligns with the requirements of the ACM0002 (Version 22.0) methodology, which states in Section 5.2.1: “If the project activity is the installation of a Greenfield power plant with or without Battery Energy Storage System (BESS), the baseline scenario is electricity delivered to the grid by the project activity that would otherwise have been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in AM TOOL 07 ‘Tool to calculate the emission factor for an electricity system (Version 07.0)’.

Justification of the Baseline Scenario:

The South African electricity grid is characterized by high carbon intensity due to its heavy reliance on coal-fired power plants, which accounts for 80% of its energy mix [14]. This high reliance on coal positions South Africa among the top greenhouse gas emitters per capita globally [15].

According to South Africa’s Integrated Resource Plan (IRP 2019), coal remains a dominant energy source, projected to account for over 50% of the electricity generation mix through 2030. The IRP outlines significant reliance on existing coal-fired power plants, with limited expansion of renewable energy capacity due to financial, regulatory, and infrastructural barriers. Despite

government efforts to increase renewable energy through initiatives like the Renewable Energy Independent Power Producer Procurement Programme (REIPPPP), the expansion pace is slow, and integration challenges persist, resulting in continued reliance on fossil fuel generation. The South African energy sector faces numerous challenges, including delays in project approvals, difficulties in securing financing, and grid connectivity issues. These barriers significantly impact the deployment of renewable energy projects, validating the assumption that the baseline scenario will continue to be dominated by fossil fuel-based power plants, including both existing and new coal and gas-fired units.

Historical data from Eskom, the primary electricity supplier, indicates a stable grid emission factor of approximately 0.96 tCO₂e/MWh over the past five years [14], driven by a heavy reliance on coal-fired plants. This trend is expected to continue in the absence of major interventions like the Redstone project, further supporting the selection of this baseline scenario. The grid emission factor for this baseline is calculated in accordance with the CDM TOOL 07 “Tool to calculate the emission factor for an electricity system” Version 07.0. Option A under the Simple OM approach was selected along with the default weighting for solar projects in the first crediting period. This results in a combined margin factor for the baseline grid of 0.96 tCO₂e/MWh. The steps taken to calculate the grid emission factor as per AM TOOL 07 are outlined in section 4.1 below. §28 – 34 of ACM0002 do not apply to this project, as shown in Table 7 below:

Table 7: Identification of Baseline scenario as per ACM0002

Section number in ACM0002	Applicable to	Applicability to the project
5.2.2	Capacity addition to an existing renewable energy power plant or integration of a BESS to an existing solar photovoltaic or wind power plant/unit	N/A – the project is a Greenfield facility and not a capacity addition to an existing renewable energy power plant. The BESS components are also greenfield and exclusive to the greenfield Umoyilanga Hybrid generation facilities.
5.2.3	Baseline scenario for retrofit or rehabilitation or replacement of an existing power plant	N/A – the project is a Greenfield facility and not a retrofit, rehabilitation or replacement of an existing power plant.
5.2.4	Baseline scenario for retrofit of an existing solar photovoltaic or wind	N/A – the project is a Greenfield facility and not a retrofit of an existing facility.

	power plant/unit with the integration of BESS	
5.2.5	Baseline scenario for pumped storage projects operating in coordination with the renewable energy plants	N/A - the project is a Greenfield facility and is not connected to a Greenfield pumped storage power plant.

The project activity is the installation of two greenfield power plants with associated BESS component. The baseline scenario is electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in TOOL 07.

3.5 Additionality

Additionality was demonstrated as stipulated in section 5.3 of ACM0002, using CDM TOOL 01 version 7. In addition, TOOL 27 version 14 and TOOL 24 version 3.1 were applied.

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

N/A – the project is located in the Republic of South Africa which is a Non-Annex 1 country and project activities are not mandated by a law, statute, or other regulatory framework.

3.5.2 Additionality Methods

Additionality is demonstrated by proving that the emission reductions achieved by the project would not have occurred in the absence of carbon finance. The following tools were used to evaluate additionality:

- CDM TOOL 01 version 7
- CDM TOOL 27 version 14
- CDM TOOL 24 version 3.1

- CDM Guidelines on the assessment of investment analysis (version 5) [16]

The input values used in the investment analysis were valid and applicable at the time of the investment decision taken by the project participant. For this project, financial close was reached on 28 November 2023, and the investment analysis was therefore carried out with values that were applicable at the time of financial close. CDM TOOL 01 version 7 prescribes a stepwise approach for evaluating additionality. These steps were followed as outlined below:

Step 0: Demonstrate whether the proposed activity is the first-of-its-kind.

Although this is one of two projects being developed in parallel that are the first to combine solar PV, wind and battery storage for electricity generation into the South African power grid, this was not considered in the additionality argument, as there are other solar PV and wind projects of a similar scale within the country.

Step 1: Identification of alternatives to the project activity

Step 1 requires that realistic and credible alternatives to the project activity be defined through the following sub-steps:

Sub-Step 1a: Define alternatives to the project activity

This step entails defining realistic and credible alternatives available to the project participants or similar project developers that provide outputs or services comparable to the proposed project activities. The technology type for this project activity was defined as a Battery Energy Storage System (BESS) supplying the South African electricity grid, charged with renewable energy.

This is shown in the table below:

Table 8: Identification of Baseline scenario as per ACM0002

Project Alternatives	Applicability to this project
Alternative a: The proposed project activity is undertaken without being registered as a VCS project	Not a Realistic scenario: The project proponent could proceed with this project without carbon finance. The project would still proceed as a BESS supplying the South African electricity grid. This is legally and practically an option, but financially it would not be a viable alternative.
Alternative b: Other realistic and credible alternative scenarios to the proposed project activity scenario that delivers electricity with comparable quality, properties and application areas, taking into account examples of scenarios	Not a realistic scenario: The supply of stable dispatchable renewable energy to the grid is made possible by the incorporation of a BESS. No other energy storage systems were deemed realistic at the project sites and scale.

identified in the underlying methodology (ACM0002).	
Alternative c: Continuation of the current situation (no project activity is undertaken.) This scenario is therefore equal to the baseline scenario where electricity is delivered to the grid through carbon-intensive fossil-fuel derived energy.	Realistic scenario: The grid could continue operating from fossil-fuel powered plants as described in the baseline scenario.

Outcome of sub-step 1a:

Alternative a: The project proponent could not proceed with this project without carbon finance.

Alternative c: Continuation of the current situation (i.e. baseline scenario).

Sub-Step 1b: Consistency with mandatory laws and regulations

The alternative(s) shall be in compliance with all mandatory applicable legal and regulatory requirements, even if these laws and regulations have objectives other than GHG reductions, e.g. to mitigate local air pollution. (This sub-step does not consider national and local policies that do not have legally binding status.)

The scenarios identified in sub-step 1a include solar, wind, and fossil-fuel derived electricity currently feeding into the South African grid. All three these technology options are in compliance with all legal and regulatory requirements in South Africa. There are no laws prohibiting the implementation of the scenarios identified, and other projects do exist in the country as mentioned in Step 0 above. In addition, all applicable environmental laws have been adhered to as indicated by the approved environmental and water use authorisations.

The applicable laws are listed in Table 6, in section 1.15 of this document.

Outcome of sub-step 1b:

The assessment indicates that scenarios listed above are in compliance with mandatory legislation and regulations in South Africa.

The project proponent can now proceed to step 2 or step 3. Project proponents may also select to complete both steps 2 and 3.

Step 2: Investment analysis

This step entails determining whether the proposed project is not:

(a) The most economically or financially attractive; or

(b) Economically or financially feasible, without the revenue from the sale of certified emission reductions (CERs).

The latest version of the “Guidelines on the assessment of investment analysis”, was also considered during this step. Option (b) was selected for analysis which is to conduct a financially attractive option. The following sub-steps were applied:

Sub-step 2a: Determine appropriate analysis method

Determine whether to apply simple cost analysis, investment comparison analysis or benchmark analysis. The scenarios identified in step 1 generate financial benefits and therefore the simple cost analysis is not suited. The alternative of continuing the current scenario does not involve an investment by the project developer and therefore the investment comparison analysis is not suited. The benchmark analysis (option III) was most suitable and therefore used for this analysis.

Sub-step 2b: Apply Investment benchmark analysis

Identify financial indicator:

According to TOOL 01, version 7, either project IRR or equity IRR can be used. The financial indicator identified as most appropriate for this type of project was identified as pre-tax project IRR. This is a parameter that is used commonly within the market. As per the CDM guidelines on the assessment of investment analysis [16], section III, a pre-tax benchmark is suited to the project IRR calculation. This is due to the impact of loan interest on income tax calculations. According to TOOL 27 §16, the applied benchmark shall be appropriate to the type of IRR calculated. Local commercial lending rates or Weighted Average Cost of Capital (WACC) are appropriate benchmarks for a project IRR. Local commercial lending rate was chosen as a suitable benchmark, as obtained from the South African Reserve Bank at the time of investment decision. At the time of investment decision, which was 28 November 2023, the commercial lending rate in South Africa was 11.75%, and therefore this value was used as pre-tax benchmark [17].

Sub-step 2c: Calculation and comparison of financial indicators

The project IRR benchmark analysis was carried out based on the financial assumptions made during the project decision making process. Based on the results, the project IRR is assessed to be less than the selected benchmark. This substantiates that the project would not be able to service debt at South Africa’s commercial lending rate and therefore the project is not financially viable without additional income such as the sale of carbon credits. Project IRR for the project activity is less than the Benchmark. Therefore, it can be concluded that the project activity is additional and is not the business-as-usual scenario.

The underlying assumptions are tabulated below:

Project Parameter	Value	Unit	Source
Total Capacity	75	MW	PPA [2]

Annual generation	384,617	MWh/yr	This figure refers to the average annual production over a 20-year period.
Deration in 1st year	0.6%	%	The expected reduction in performance over time as per the production model
Deration 2nd year onward	0.7%	%	The expected reduction in performance over time as per the production model
Total plant Capex This includes infrastructure, general and electrical equipment, mechanical equipment, civil works and construction costs, site office establishment, instrumentation and control, environmental approvals, other pre-operative expenses	6,972,760	ZAR	Project Financials (Approved financial model as at time of financial close)
Tariff rate	1,690	ZAR/MWh	PPA (weighted average)[2]
O&M Costs including Energy management system costs, preventative maintenance on equipment, corrective maintenance, staff and labour costs, infrastructure management, Health, safety and environmental compliance.	123,578	kZAR/yr	Average over 20 years. Cost estimations made by project developer. A portion of the contract fees are in Euro, a portion in USD and a portion in ZAR. Each amount was converted to ZAR at the exchange rate at time of financial decision, and inflated annually by South Africa's projected core inflation rate (CIR) at time of investment decision.
ZAR/USD Exchange rate	18.32	ZAR/\$	Rand/Dollar exchange rate at time of investment decision
ZAR/Euro Exchange rate	19.93	ZAR/€	Rand/Euro exchange rate at time of investment decision
Escalation	4.5	%	South Africa's projected core inflation rate at time of investment decision [18]

Insurance	19,000	kZAR/yr	Supplier contracts, 1 st year, inflated annually by projected CIR at time of investment decision
Salvage value	27,953	kZAR	Decommissioning report
Commercial lending rate at time of investment decision	11.75	%	Released by the SA Reserve Bank on a monthly basis [17]

Based on the assumptions above, the pre-tax project IRR would be 8.06% which is below the benchmark of 11.75%

Sub-step 2d: Sensitivity Analysis

A sensitivity analysis was conducted to demonstrate that the project IRR value remains below the benchmark threshold, even when there are variations to the critical input assumptions, as required by the Tool. A range of +10% and -10% was applied. The following variables are considered in the sensitivity analysis:

- Capital cost
- Operations and maintenance costs
- Electricity tariff
- Annual generation

Parameter	Project IRR at -10%	Project IRR at +- 0%	Project IRR at + 10%
Capex	9.32%	8.06%	6.97%
O&M	8.34%	8.06%	7.77%
Tariff	6.51%	8.06%	9.51%
Annual generation	6.51%	8.06%	9.51%

The results of the sensitivity analysis indicate that none of the input variances put the project at risk of exceeding the benchmark project IRR, which is the commercial lending rate of 11.75%.

In conclusion, the proposed project activity requires additional revenues from the sale of carbon credits to be economically attractive to project developers and surpass or get closer to the benchmark.

Step 3: Barrier analysis

§28 of TOOL 01 allows for project proponents to proceed to either Step 2 (Investment analysis) or Step 3 (Barrier analysis). Project proponents may also select to complete both Steps 2 and 3. Accordingly, Step

2 is selected. Step 3 Barrier analysis was not deemed necessary and was not considered in this analysis in accordance with the requirements of the Tool.

Step 4: Common practice analysis

The common practice analysis was carried out as stipulated in TOOL 24 version 3.1. This tool outlines a step wise approach, which is described below:

Step 1: Calculate applicable capacity output range as +/- 50% of the total design capacity.

The dispatch capacity for the Umoyilanga project is 75 MW and therefore the capacity range for this analysis ranges from 37.5 MW to 112.5 MW.

Step 2: Identify similar projects (both VCS and non-VCS) which fulfill all of the following conditions:

Criteria	Applicability to this project	Projects identified
The projects are located in the applicable geographical area;	The applicable area is the geographical borders of South Africa.	All the renewable energy projects listed were considered for this analysis.
The projects apply the same measure as the proposed project activity;	The measure applied is renewable energy fed into the South African electricity grid.	5,729.7 MW of renewable energy is supplied to the South African grid (extracted from Eskom's website) [19].
The projects use the same energy source/fuel and feedstock as the proposed project activity, if a technology switch measure is implemented by the proposed project activity;	The fuel source is a combination of solar and wind with integrated battery storage managed by a virtual power plant controller.	There is only one similar project meeting this criteria in South Africa and it is currently under development [20].
The plants in which the projects are implemented produce goods or services with comparable quality, properties and applications areas (e.g. clinker) as the proposed project plant;	Renewable electricity	Solar PV and Wind Energy with BESS
The capacity or output of the projects is within the applicable capacity or output range calculated in Step 1;	37.5 MW to 112.5 MW	The capacity of the other project is 128 MW and therefore it falls outside the scope N _{all} = 0

The projects started commercial operation before the project design document (CDM-PDD) is published for global stakeholder consultation or before the start date of proposed project activity, whichever is earlier for the proposed project activity.	There are no comparable, operational projects within South Africa at the time of publishing this PD for global stakeholder consultation.	The other project is not yet commissioned and therefore not valid.
--	--	--

It can therefore be concluded that $N_{ALL} = 0$.

Steps 4 and 5 of this section are not applicable since $N_{ALL} = 0$

It can therefore be concluded that the proposed project activity is not common practice within South Africa.

3.6 Methodology Deviations

There are no methodology deviations applied in this project.

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

The baseline emissions were quantified in accordance with the methodology ACM0002 version 22.0. For this project, it includes electricity delivered to the South African National grid. This was calculated as set out in AM TOOL 07, and the calculation steps are described below.

Baseline Emission Calculation

Baseline emissions include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Equation 1 (ACM0002 Equation 11)

Where:

BE_y	=	Baseline emissions in year y (tCO ₂ /yr)
$EG_{PJ,y}$	=	Net electricity fed into the grid supplied by the project activity in year y (MWh/yr)
$EF_{grid,CM,y}$	=	Combined margin CO ₂ emission factor for grid connected power generation in year y calculated using TOOL 07 (tCO ₂ /MWh)

The results obtained are: 369,232 tCO₂/annum, as shown below:

Parameter name	Value	Unit
BE_y	369,232	tCO ₂ / yr
$EG_{PJ,y}$	384,617	MWh/yr
$EF_{grid,CM,y}$	0.96	tCO ₂ /MWh

The project is a greenfield power plant and therefore $EG_{PJ,y}$ is calculated as follows:

$$EG_{PJ,y} = EG_{facility,y}$$

Equation 2 (ACM0002 Equation 12)

Where:

$EG_{PJ,y}$ = Net electricity generation fed into the grid as a result of the implementation of the project activity in year y (MWh/yr)

$EG_{facility,y}$ = Net electricity generation supplied by the project plant to the grid in year y (MWh/yr)

The results obtained are: 384,617 MWh/yr (from technical specifications)

Grid Emission Factor Calculation

The grid emission factor for South Africa's national electricity grid was calculated as the weighted average of the system's operating margin (OM) and build margin (BM). The operating margin refers to the group of existing power plants whose current electricity generation would be affected by the proposed renewable energy project activity. The build margin refers to the group of prospective power plants whose construction and future operation would be affected by the proposed renewable energy project activity. This calculation is only applicable to grid-connected power plants and does not include off-grid power plants.

In accordance with TOOL 07 The following 6 steps are followed:

- **Step 1:** Identify the relevant electricity systems
- **Step 2:** Choose whether to include off-grid power plants in the project electricity system (optional)
- **Step 3:** Select a method to determine the operating margin (OM)
- **Step 4:** Calculate the operating margin emission factor according to the selected method
- **Step 5:** Calculate the build margin (BM) emission factor
- **Step 6:** Calculate the combined margin (CM) emission factor.

4.1.1 Step 1: Identify the project electricity system

The project electricity system was identified, using Option 2, as: The national grid of South Africa – administered by the national power utility, Eskom Holding SOC Ltd. This is shown schematically in Figure 12 below.

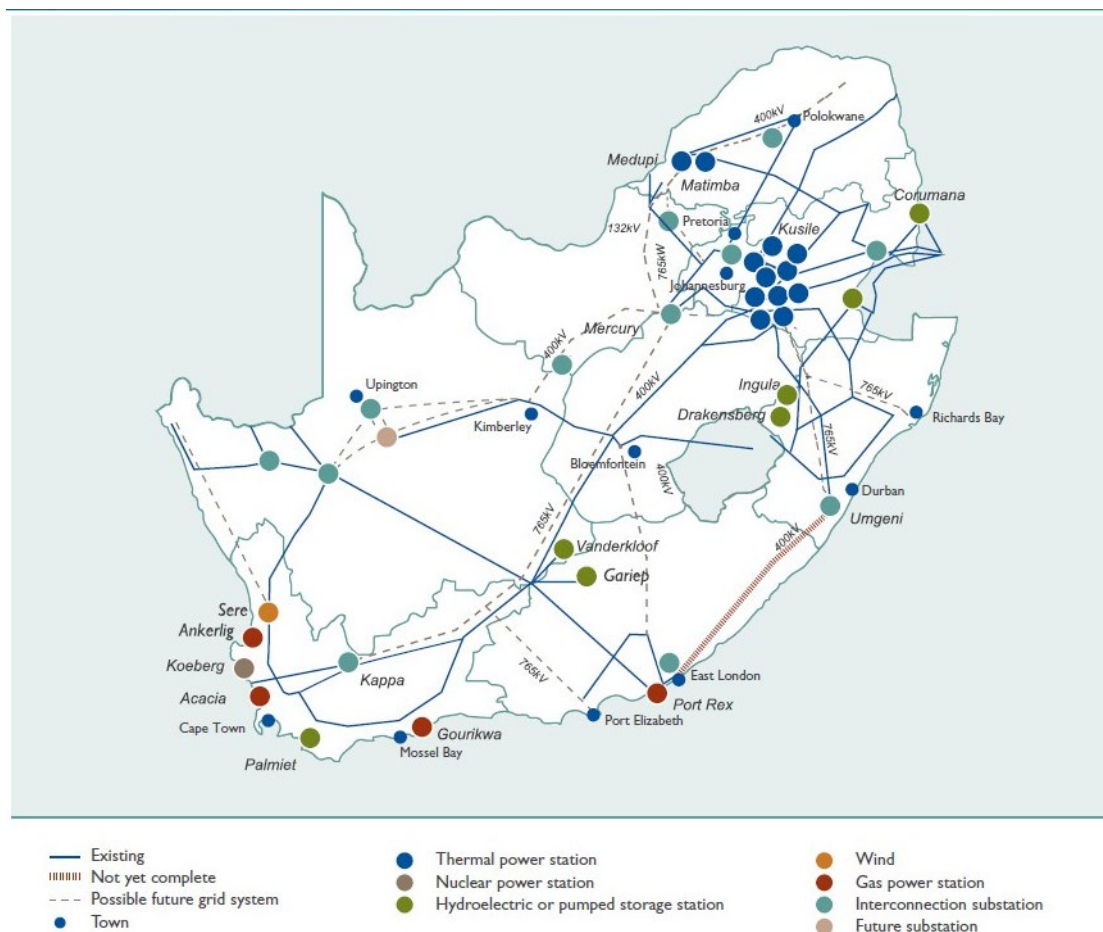


Figure 7: Project electricity system

South Africa's electricity grid is not connected to an electricity system located partially or totally in an Annex I country, therefore the grid emission factor is not zero.

4.1.2 Step 2: Choose whether to include off-grid power plants in the project electricity system

Only grid connected plants (shown in Table 10). were included in this calculation.

4.1.3 Step 3: Operating Margin method selection

The operating margin (OM) emission factor ($EF_{grid,OM,y}$) was calculated based on the simple OM method (§38.a in TOOL 07). This was decided based on the following criteria:

- The share of electricity supplied by Low-Cost/Must-Run (LCMR) plants in the last 5 years has ranged from 19% – 23%. It has been consistently lower than 50%, as calculated using TOOL 07 40(a) Approach 1, by Equation 3 below:

$$Share_{LCMR} = average \left[\frac{EG_{LCMR(y-4)}}{total_{y-4}}, \dots, \frac{EG_{LCMR(y)}}{total_y} \right]$$

Equation 3 (TOOL 07, 40 Equation 1)

Where:

$Share_{LCMR}$ = Share of the low cost/must run resources (%)

EG_{LCMRy} = Electricity generation supplied by LCMR plants in year y (MWh)

$total_y$ = Total electricity supplied to the project electricity system in year y (MWh)

Y = The most recent year for which data is available

The ex-ante option (TOOL 07 §42(a)) was chosen to determine the simple OM. The 3-year generation-weighted average emission factor was calculated based on the most recent data available at the time of listing the project on Verra's registry. For this project, the most recent data available is 2023. Therefore, the generation weighted average of 2021, 2022, and 2023 was used.

The results obtained are shown in Table 8 below

Table 9: LCMR Analysis

Parameter name	Value	Unit
$Share_{LCMR}$: 2023	23.1%	Per cent
$Share_{LCMR}$: 2022	21.3%	Per cent
$Share_{LCMR}$: 2021	19.4%	Per cent
$Share_{LCMR}$: Average (2021 – 2023)	21.2%	Per cent
EG_{LCMRy} : 2023	49,691	GWh/yr
EG_{LCMRy} : 2022	48,092	GWh/yr
EG_{LCMRy} : 2021	42,495	GWh/yr
$total_y$: 2023	215,318	GWh/yr
$total_y$: 2022	226,226	GWh/yr
$total_y$: 2021	219,423	GWh/yr

4.1.4 Step 4: Operating margin calculation

The simple OM emission factor was calculated as the generation-weighted average CO₂ emissions per unit net electricity generation (tCO₂/MWh) of all generating power plants serving the system, not including LCMR power plants. The following plants were identified as LCMR, and therefore excluded from the OM calculation:

Table 10 LCMR plants serving the project electricity system

LCMR Plant name	Technology	Reason for classification as LCMR
• Gariep • Vanderkloof • Mbashe • First Falls • Ncora • Second Falls	Hydroelectric power stations	Renewable (low generation costs)
• Drakensberg • Ingula • Palmiet	Pumped storage stations	Only used during periods of peak demand (must run)
• Acacia • Ankerlig • Gourikwa • Port Rex	Gas and liquid turbine peaking stations	Operated only during hours of highest daily, weekly or seasonal loads (must run)
• Sere	Wind energy stations	Renewable (low generation costs)
• Koeberg	Nuclear power stations	Nuclear listed as LCMR
• Biomass • Concentrated solar power • Gas/liquid fuel • Hydroelectric • Landfill • Solar PV • Wind	Imports and REIPPP plants	Imports or renewable

The power plants used in the OM calculation are shown in Table 10 below. The simple OM was calculated based on the net electricity generation and the CO₂ emission factor of each power unit (TOOL 07 paragraph 47, option A), as per Equation 4.

$$EF_{grid,OMsimple,y} = \frac{\sum m EG_{m,y} \times EF_{EL,m,y}}{\sum m EG_{m,y}}$$

Equation 4 (TOOL 07, 48 Equation 3)

Where:

$EF_{grid,OMsimple,y}$	=	Simple operating margin CO ₂ emission factor in year y (tCO ₂ /MWh)
$EG_{m,y}$	=	Net electricity delivered to the grid by power unit m in year y (MWh)
$EF_{EL,m,y}$	=	CO ₂ emission factor of power unit m in year y (tCO ₂ /MWh)
m	=	All power units serving the grid in year y except low-cost/must-run power units
y	=	The relevant year as per the data vintage chosen

The results obtained are: $EF_{grid,OMsimple,y} = 1.004$ tCO₂/MWh

All values used are shown in Table 10.

Table 11: Operating Margin Plant Data

Plant Name	Electricity generated (MWh/yr) 2021	Electricity generated (MWh/yr) 2022	Electricity generated (MWh/yr) 2023	Emission factor 2023 (tCO ₂ /MWh)	Emission factor 2022 (tCO ₂ /MWh)	Emission factor 2021 (tCO ₂ /MWh)
Arnot	10,148,766	9,085,064	7,530,158	1.17	1.16	1.14
Duvha	11,230,400	9,822,489	9,555,266	1.11	1.13	1.13
Hendrina	4,873,506	4,738,179	2,136,620	1.35	1.24	1.20
Kendal	15,106,544	15,139,746	15,750,630	1.02	1.05	1.04
Kriel	11,328,488	12,597,519	10,727,877	1.32	1.28	1.22
Lethabo	21,832,979	22,890,481	21,367,191	1.00	0.97	0.95
Matimba	23,533,126	23,180,336	23,257,630	1.02	1.04	0.99
Majuba	21,857,835	19,823,301	15,540,628	1.19	1.13	1.09
Matla	18,560,297	18,412,664	15,170,449	1.30	1.27	0.99
Tutuka	10,554,111	8,524,715	7,564,337	1.18	1.23	1.22
Camden	2,518,438	6,670,341	7,378,745	1.10	1.16	1.25
Grootvlei	2,992,859	2,445,162	2,416,293	1.21	1.27	1.24
Komati	708,862	608,728	369,330	1.01	1.03	1.06
Kusile	8,553,383	7,992,189	9,119,476	1.01	1.30	1.09
Medupi	19,753,895	22,637,675	23,246,212	0.88	0.92	0.91
Total	183,553,489	184,568,589	171,130,842			

The emission factor of each power unit ($EF_{EL,m,y}$) was determined following option A1 (paragraph 49.a) as shown in Equation 5 below:

$$EF_{EL,m,y} = \frac{\sum_i FC_{i,m,y} \times NCV_{i,y} \times EF_{CO2,i,y}}{\sum EG_{m,y}}$$

Equation 5 (TOOL 07, 49.(a) Equation 4)

Where:

$EF_{EL,m,y}$	=	CO ₂ emission factor of power unit m in year y (tCO ₂ /MWh)
$FC_{i,m,y}$	=	Amount of fuel type i consumed by power unit m in year y (Mass or volume unit)
$NCV_{i,y}$	=	Net calorific value of fuel type i in year y (GJ/mass or volume unit)
$EF_{CO2,i,y}$	=	CO ₂ emission factor of fuel type i in year y (tCO ₂ /GJ)
$EG_{m,y}$	=	Net quantity of electricity delivered to the grid by power unit m in year y (MWh)
m	=	All power units serving the grid in year y except low-cost/must-run power units
i	=	All fuel types combusted in power unit m in year y
y	=	The relevant year as per the data vintage chosen

The emission factor for each power unit ($EF_{EL,m,y}$) is shown in Table 10 above.

The values used in Equation 5 are shown in Table 11 below.

$EG_{m,y}$ was obtained from Eskom's website as well as their annual Integrated Reports.

Table 12: Parameters for plant emission factor calculation

Plant Name	Coal usage (t) 2021	Coal usage (t) 2022	Coal usage (t) 2023	Fuel oil usage (t) 2023	Fuel oil usage (t) 2022	Fuel oil usage (t) 2021	Net calorific value coal (GJ/t)	Net calorific value fuel oil (GJ/t)	Emission factor coal (tCO ₂ /GJ)	Emission factor fuel oil (tCO ₂ /GJ)
Arnot	5,663,126	5,186,766	4,293,316	45,179.7	37,792.2	36,433.7	21.43	40.40	0.09	0.08
Duvha	6,581,000	5,732,195	5,428,210	71,179.4	39,550.9	34,435.1	20.39	40.40	0.09	0.08
Hendrina	3,024,527	3,024,981	1,486,626	7,722.1	11,012.5	7,013.8	20.51	40.40	0.09	0.08
Kendal	9,272,316	9,362,475	9,395,998	106,402.9	61,174.9	32,683.4	17.87	40.40	0.09	0.08
Kriel	6,694,760	7,838,445	6,865,127	53,949.1	39,516.2	47,877.9	21.64	40.40	0.09	0.08
Lethabo	15,158,029	16,277,939	15,646,686	14,075.3	12,792.0	9,740.7	14.53	40.40	0.09	0.08
Matimba	12,938,911	13,430,751	13,179,578	14,066.5	9,944.3	13,465.6	19.15	40.40	0.09	0.08
Majuba	12,773,092	11,924,693	9,831,151	140,169.2	149,002.7	89,746.1	19.58	40.40	0.09	0.08
Matla	10,694,976	10,842,908	9,536,046	15,923.0	14,520.0	9,179.0	18.19	40.40	0.09	0.08
Tutuka	6,697,281	5,510,214	4,697,013	64,592.8	68,779.3	124,521.3	19.83	40.40	0.09	0.08
Camden	1,569,620	3,932,224	4,129,344	49,016.8	57,014.4	41,321.3	20.45	40.40	0.09	0.08
Grootvlei	1,938,931	1,544,269	1,518,602	21,199.1	61,318.1	19,380.3	20.01	40.40	0.09	0.08
Komati	402,289	335,833	197,954	1,773.0	1,332.0	2,214.0	19.72	40.40	0.09	0.08
Kusile	5,047,912	4,783,403	5,035,930	10,351.0	534,043.2	32,165.0	19.42	40.40	0.09	0.08
Medupi	9,763,379	11,351,788	11,181,866	23,246.2	22,637.7	19,753.9	19.42	40.40	0.09	0.08

4.1.5 Step 5: Calculate the build margin (BM) emission factor

An ex-ante approach (option 1 in TOOL 07 §72(a)) is selected to calculate the BM emission factor. The most recent information available on units already built for sample group m at the time of submitting the project for validation was used. The procedure followed to determine the sample group of power units (m) used to calculate the build margin is described below. The total electricity generation (AEG_{total}) of all power plants directly supplying the system in 2023 was 184,165.63 GWh.

The set of five power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently ($SET_{5 \text{ units}}$) was identified as shown in Table 12. The annual generation of $SET_{5 \text{ units}}$ was 35,365.31 GWh in 2023, and this set comprises 19.2% of AEG_{total} .

Table 13: Five most Recent Plants supplying the grid

No.	Plant name	Commissioning Date	Electricity supplied UOM	2023
1	Kusile	2017/08/30	GWh/yr	9,119.48
2	Medupi	2015/08/22	GWh/yr	23,246.21
3	Grootvlei	2008/04/30	GWh/yr	2,416.29
4	Komati	2009/03/31	GWh/yr	369.33
5	Sere Wind	2015/03/01	GWh/yr	214.00
TOTAL				35,365.31

The set of power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently and that comprise 20 per cent of AEG_{total} was determined next, and $SET_{\geq 20\%}$ generated 38,379.02 MWh as shown in Table 13 below, comprised 23.4% of AEG_{total} .

Table 14: Set of power plants that contribute 20% to the Eskom fleet power generation

No.	Plant name	Commissioning Date	UoM	Power Produced in Year: 2023
1	Kusile	2017/08/30	GWh/yr	9,119.48
2	Medupi	2015/08/22	GWh/yr	23,246.21
3	Ankerlig	2007/03/01	GWh/yr	1,332.49
4	Gourikwa	2007/07/01	GWh/yr	1,681.22
5	Grootvlei	2008/04/30	GWh/yr	2,416.29
6	Komati	2009/03/31	GWh/yr	369.33
7	Sere Wind	2015/03/01	GWh/yr	214.00
TOTAL				38,379.02

SET>20% comprised the larger annual electricity generation and therefore forms the sample set SET_{sample}, but it did contain plants that were commissioned more than 10 years ago and therefore step (d) is followed. Units that were commissioned more than 10 years ago were excluded from the set (those shaded grey in Table 13), and CDM, Gold Standard (GS), and VCS projects were included until the electricity generation of the new set comprises 20% of the annual electricity generation of the project electricity system. This resulted in a set that comprised 20% of the project electricity system and therefore steps (e) and (f) were not required. The resulting SET_{sample-CDM} group of power units, m, used to calculate the build margin is shown

Table 15: Plant set for BM Calculation

No.	Plant name	Commissioning Date	UoM	Power Produced in Year: 2023
Eskom Fleet Plants				
1	Kusile	2017/08/30	GWh/yr	9,119.48
2	Medupi	2015/08/22	GWh/yr	23,246.21
3	Sere Wind	2015/03/01	GWh/yr	214.00
	Total Eskom			32,579.69
CDM, VCS and Gold Standard Plants				
4	Amakhala Emoyeni Grid Connected 138.6 MW Wind Farm, Phase 1	2013/03/01	GWh/yr	375.17
5	Karoo Renewable Energy Facility (Nobelsfontein Solar PV)	2014/04/01	GWh/yr	94.53
6	Bokpoort CSP (Concentrating Solar Power) Project, South Africa	2013/01/10	GWh/yr	225.86
7	Springbok Grid Connected 55.5 MW Wind Farm (Longyuan Mulilo)	2013/06/01	GWh/yr	130.30
8	Electricawinds 30 MW Wind Project at Riverbank	2013/12/01	GWh/yr	80.30
9	Longyuan Mulilo 1	2012/01/01	GWh/yr	308.14
10	Longyuan Mulilo 2 de Aar, 144 MW	2012/01/01	GWh/yr	497.38
11	Karoo Renewable Energy Facility (Nobelsfontein Wind)	2012/12/01	GWh/yr	878.44
12	Caledon Wind Farm in South Africa	2012/06/01	GWh/yr	406.01
13	Langa Energy Photovoltaic Solar Energy Facility	2012/03/05	GWh/yr	140.45
14	Indwe Wind Project	2012/06/01	GWh/yr	148.09
15	Grid Connected Wind Power Plant in Witberg	2012/02/01	GWh/yr	507.20
16	Red Cap Kouga Wind Farm	2011/11/02	GWh/yr	290.00
	TOTAL CDM, VCS and GS			4,081.85
	Total			36,661.5

As the annual electricity generation of this set comprises at least 20% of the annual electricity generation of the project electricity system (i.e. $AEG_{SET-sample-CDM} > 0.2 \times AEG_{total}$), this sample group ($SET_{sample-CDM}$) is used to calculate the build margin.

The build margin emission factor was calculated as the generation-weighted average emission factor (tCO_2/MWh) for power units m during 2023, which is the most recent year y for which electricity generation data is available, using Equation 6:

$$EF_{grid,BM,y} = \frac{\sum m EG_{m,y} \times EF_{EL,m,y}}{\sum m EG_{m,y}}$$

Equation 6 (TOOL 07, 77 Equation 15)

Where:

$EF_{grid,BM,y}$	=	Build margin CO_2 emission factor in year y (tCO_2/MWh)
$EG_{m,y}$	=	Net quantity of electricity delivered to the grid by power unit m in year y (MWh)
$EF_{EL,m,y}$	=	CO_2 emission factor of power unit m in year y (tCO_2/MWh)
m	=	Power units included in the build margin
y	=	Most recent historical year for which electricity generation data is available

The results obtained are: $EF_{grid,BM,y} = 0.81$

$EG_{m,y}$ and $EF_{EL,m,y}$ were used as shown in Table 10, $EG_{m,y}$ was used as shown in Table 14.

4.1.6 Step 6: Calculate the combined margin (CM) emission factor

The combined margin (CM) emission factor ($EF_{grid,CM,y}$) was calculated as the weighted average CM, as per Equation 7.

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times w_{OM} + EF_{grid,BM,y} \times w_{BM}$$

Equation 7 (TOOL 07, 85 Equation 16)

As per TOOL I07, the following weighted values (for solar power generation) were used for w_{OM} and w_{BM} :

- $w_{OM} = 0.75$
- $w_{BM} = 0.25$

The results obtained are: $EF_{grid,CM,y} = 0.96$

4.2 Project Emissions

Project emissions are calculated as follows, in accordance with ACM0002 version

$$PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y}$$

Equation 8 (ACM0002 Equation 1)

Where:

PE_y = Project emissions in year y (tCO₂)

$PE_{FF,y}$ = Project emissions from fossil fuel consumption in year y (tCO₂)

$PE_{GP,y}$ = Project emissions from the operation of geothermal power plants due to the release of non-condensable gases in year y (tCO₂e/yr)

$PE_{HP,y}$ = Project emissions from water reservoirs of hydro power plants (tCO₂e/ yr)

The project does not involve geothermal power plants or water reservoir of hydro power plants therefore the equation simplifies to:

$$PE_y = PE_{FF,y}$$

This plant will not use fossil fuels for electricity generation, only for emergency backup, therefore $PE_{FF,y}$ can be neglected, in accordance with ACM0002 paragraph 42. Thus project emissions are zero.

4.3 Leakage Emissions

No leakage emissions are considered. The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g. extraction, processing, transport etc.) are neglected, in accordance with ACM0002.

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

The project's emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y$$

Equation 9 (ACM0002 Equation 17)

Where:

ER_y = Emission reductions in year y (tCO₂/yr)

BE_y = Baseline emissions in year y (tCO₂/yr)

PE_y = Project emissions in year y (tCO₂/yr)

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCUs (tCO ₂ e)	Estimated removal VCUs (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
15-October-2025 to 31-December-2025	76 923	0	0	76 923	0	76 923
01-January-2026 to 31-December-2026	369,232	0	0	369,232	0	369,232
01-January-2027 to 31-December-2027	369,232	0	0	369,232	0	369,232
01-January-2028 to 31-December-2028	369,232	0	0	369,232	0	369,232
01-January-2029 to 31-December-2029	369,232	0	0	369,232	0	369,232
01-January-2030 to 31-December-2030	369,232	0	0	369,232	0	369,232
01-January-2031 to 31-December-2031	369,232	0	0	369,232	0	369,232

01-January-2032 to 14-October-2032	292 309	0	0	292 309	0	292 309
Total	2,584,626	0	0	2,584,626	0	2,584,626

5 MONITORING

5.1 Data and Parameters Available at Validation

The table below provides all data and parameters that are determined or available at validation and remain fixed throughout the project crediting period.

Data / Parameter	$EF_{grid,OM,y}$
Data unit	tCO ₂ /MWh
Description	Simple operating margin CO ₂ emission factor in year y
Source of data	Calculated using Equation 1, as per TOOL 07 “Tool to calculate the emission factor for an electricity system” Data Sources: Records published on Eskom’s website monthly for each power station to fulfil the Atmospheric emission license (EAL) requirements. Eskom’s integrated annual report for 2023
Value applied	1.004
Justification of choice of data or description of measurement methods and procedures applied	The simple OM emission factor was calculated as the generation-weighted average CO₂ emissions per unit net electricity generation (tCO₂/MWh) of all generating power plants serving the system, not including low-cost/must-run power plants/units. This means that only Eskom’s coal-based power plants were included in the calculation. The OM was calculated as per option A (paragraph 47.a) of TOOL 07 “Tool to calculate the emission factor for an electricity system”, based on the net electricity generation and a CO₂ emission factor of each power unit.
Purpose of data	Calculation of baseline emissions
Comments	No additional comments

Data / Parameter	$EF_{grid,BM,y}$
Data unit	tCO ₂ /MWh
Description	Build margin CO ₂ emission factor in year y

Source of data	Calculated using Equation 6, as per TOOL 07 “Tool to calculate the emission factor for an electricity system” Data source:
Value applied	0.81
Justification of choice of data or description of measurement methods and procedures applied	The BM was calculated as per option 1 (paragraph 72) of TOOL 07 “Tool to calculate the emission factor for an electricity system”. An ex-ante approach was followed, based on the most recent information available on units already built. The BM will be updated for the second crediting period based on the most recent information available at that time. Capacity additions from retrofits of power plants are not included in the calculation of the build margin emission factor.
Purpose of data	Calculation of baseline emissions
Comments	No additional comments

Data / Parameter	$EF_{grid,CM,y}$
Data unit	tCO ₂ /MWh
Description	Combine margin CO ₂ emission factor in year y
Source of data	Calculated using Equation 7, as per TOOL 07 “Tool to calculate the emission factor for an electricity system” Data Source:
Value applied	0.96
Justification of choice of data or description of measurement methods and procedures applied	The combined margin (CM) emission factor ($EF_{grid,CM,y}$) was calculated based on the weighted average CM method (paragraph 81.a) since the data to determine the BM is available. This was done as per equation 16 of TOOL 07 “Tool to calculate the emission factor for an electricity system”. Where WOM is equal to: 0.75 and WBM is equal to: 0.25, for the first crediting period and for subsequent crediting periods
Purpose of data	Calculation of baseline emissions
Comments	No additional comments

Data / Parameter	Share _{LCMR}						
Data unit	Percent (%)						
Description	Share of the low cost/must run resources						
Source of data	Determined from electricity generation data reported in Eskom’s annual Integrated Report for 2023 [14] and records published on Eskom’s website [21] for each power station.						
Value applied	Annual LCMR (%) <table><tr><td>2021</td><td>2022</td><td>2023</td></tr><tr><td>19.4%</td><td>21.3%</td><td>23.1%</td></tr></table>	2021	2022	2023	19.4%	21.3%	23.1%
2021	2022	2023					
19.4%	21.3%	23.1%					
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa’s national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 8						
Purpose of data	To calculate baseline emissions, using Equation 3.						
Comments	The simple operating margin (OM) can be used where low-cost/must-run resources constitute less than 50 per cent of total grid generation (excluding electricity generated by off-grid power plants) in the average of the five most recent years, determined by using Equation 1 under Approach 1 from section 40 in TOOL 07.						

Data / Parameter	EG _{LCMR,y}						
Data unit	MWh						
Description	Electricity generation supplied to the project electricity system by the low cost/must run resources in year y						
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 and records published on Eskom's website [22] for each power station.						
Value applied	<table border="1"> <tr> <td>EG_{LCMRy}: 2023</td><td>49,691,000</td></tr> <tr> <td>EG_{LCMRy}: 2022</td><td>48,092,000</td></tr> <tr> <td>EG_{LCMRy}: 2021</td><td>42,495,000</td></tr> </table>	EG _{LCMRy} : 2023	49,691,000	EG _{LCMRy} : 2022	48,092,000	EG _{LCMRy} : 2021	42,495,000
EG _{LCMRy} : 2023	49,691,000						
EG _{LCMRy} : 2022	48,092,000						
EG _{LCMRy} : 2021	42,495,000						
Justification of choice of data or description of	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from						

measurement methods and procedures applied	Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 8
Purpose of data	To calculate baseline emissions, using Equation 3.
Comments	The simple operating margin (OM) can be used where low-cost/must-run resources constitute less than 50 per cent of total grid generation (excluding electricity generated by off-grid power plants) in the average of the five most recent years, determined by using Equation 1 under Approach 1 from section 40 in TOOL 07.

Data / Parameter	$total_y$		
Data unit	MWh		
Description	Total electricity generation supplied to the project electricity system in year y		
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 and monthly records published on Eskom's website [23] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [14] and records published on Eskom's website [21] for each power station		
Value applied	$total_y$: 2023	215,318	GWh/yr
	$total_y$: 2022	226,226	GWh/yr
	$total_y$: 2021	219,423	GWh/yr
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 8As presented in Table 8		
Purpose of data	To calculate baseline emissions, using Equation 3.		
Comments	The simple operating margin (OM) can be used where low-cost/must-run resources constitute less than 50 per cent of total grid generation (excluding electricity generated by off-grid power plants) in the average of the five most recent years, determined by using Equation 1 under Approach 1 from section 40 in TOOL 07.		

Data / Parameter	Y
Data unit	Year

Description	The most recent year for which data is available
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [14] and monthly records published on Eskom's website [23] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [14] and records published on Eskom's website [21] for each power station
Value applied	2023
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 8
Purpose of data	To calculate baseline emissions, using Equation 3.
Comments	The simple operating margin (OM) can be used where low-cost/must-run resources constitute less than 50 per cent of total grid generation (excluding electricity generated by off-grid power plants) in the average of the five most recent years, determined by using Equation 1 under Approach 1 from section 40 in TOOL 07.

Data / Parameter	$EF_{grid,OMsimple,y}$
Data unit	tCO ₂ e/MWh
Description	Simple operating margin CO ₂ emission factor in year y
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (AEL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [14] and records published on Eskom's website [21] for each power station
Value applied	1.004
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 10
Purpose of data	To calculate baseline emissions, using Equation 4.

Comments	The simple OM emission factor is calculated as the generation-weighted average CO ₂ emissions per unit net electricity generation (t CO ₂ /MWh) of all generating power plants serving the system, not including low-cost/must-run power plants. using Equation 3 under Option A from section 48 in TOOL 07.																																																																							
Data / Parameter	EG _{m,y}																																																																							
Data unit	MWh																																																																							
Description	Net electricity delivered to the grid by power unit m in year y																																																																							
Source of data	Determined from electricity generation data reported in Eskom’s annual Integrated Report for 2023 [11] and monthly records published on Eskom’s website [14] for each power station to fulfil the Atmospheric emission license (AEL) requirements, and from records published on Eskom’s website [21] for each power station																																																																							
Value applied	As per Table 10 <table><tr><th>Plant Name</th><th>Electricity generated (MWh/yr) 2021</th><th>Electricity generated (MWh/yr) 2022</th><th>Electricity generated (MWh/yr) 2023</th></tr><tr><td>Arnot</td><td>10,148,766</td><td>9,085,064</td><td>7,530,158</td></tr><tr><td>Duvha</td><td>11,230,400</td><td>9,822,489</td><td>9,555,266</td></tr><tr><td>Hendrina</td><td>4,873,506</td><td>4,738,179</td><td>2,136,620</td></tr><tr><td>Kendal</td><td>15,106,544</td><td>15,139,746</td><td>15,750,630</td></tr><tr><td>Kriel</td><td>11,328,488</td><td>12,597,519</td><td>10,727,877</td></tr><tr><td>Lethabo</td><td>21,832,979</td><td>22,890,481</td><td>21,367,191</td></tr><tr><td>Matimba</td><td>23,533,126</td><td>23,180,336</td><td>23,257,630</td></tr><tr><td>Majuba</td><td>21,857,835</td><td>19,823,301</td><td>15,540,628</td></tr><tr><td>Matla</td><td>18,560,297</td><td>18,412,664</td><td>15,170,449</td></tr><tr><td>Tutuka</td><td>10,554,111</td><td>8,524,715</td><td>7,564,337</td></tr><tr><td>Camden</td><td>2,518,438</td><td>6,670,341</td><td>7,378,745</td></tr><tr><td>Grootvlei</td><td>2,992,859</td><td>2,445,162</td><td>2,416,293</td></tr><tr><td>Komati</td><td>708,862</td><td>608,728</td><td>369,330</td></tr><tr><td>Kusile</td><td>8,553,383</td><td>7,992,189</td><td>9,119,476</td></tr><tr><td>Medupi</td><td>19,753,895</td><td>22,637,675</td><td>23,246,212</td></tr><tr><td>Total</td><td>183,553,489</td><td>184,568,589</td><td>171,130,842</td></tr></table>				Plant Name	Electricity generated (MWh/yr) 2021	Electricity generated (MWh/yr) 2022	Electricity generated (MWh/yr) 2023	Arnot	10,148,766	9,085,064	7,530,158	Duvha	11,230,400	9,822,489	9,555,266	Hendrina	4,873,506	4,738,179	2,136,620	Kendal	15,106,544	15,139,746	15,750,630	Kriel	11,328,488	12,597,519	10,727,877	Lethabo	21,832,979	22,890,481	21,367,191	Matimba	23,533,126	23,180,336	23,257,630	Majuba	21,857,835	19,823,301	15,540,628	Matla	18,560,297	18,412,664	15,170,449	Tutuka	10,554,111	8,524,715	7,564,337	Camden	2,518,438	6,670,341	7,378,745	Grootvlei	2,992,859	2,445,162	2,416,293	Komati	708,862	608,728	369,330	Kusile	8,553,383	7,992,189	9,119,476	Medupi	19,753,895	22,637,675	23,246,212	Total	183,553,489	184,568,589	171,130,842
Plant Name	Electricity generated (MWh/yr) 2021	Electricity generated (MWh/yr) 2022	Electricity generated (MWh/yr) 2023																																																																					
Arnot	10,148,766	9,085,064	7,530,158																																																																					
Duvha	11,230,400	9,822,489	9,555,266																																																																					
Hendrina	4,873,506	4,738,179	2,136,620																																																																					
Kendal	15,106,544	15,139,746	15,750,630																																																																					
Kriel	11,328,488	12,597,519	10,727,877																																																																					
Lethabo	21,832,979	22,890,481	21,367,191																																																																					
Matimba	23,533,126	23,180,336	23,257,630																																																																					
Majuba	21,857,835	19,823,301	15,540,628																																																																					
Matla	18,560,297	18,412,664	15,170,449																																																																					
Tutuka	10,554,111	8,524,715	7,564,337																																																																					
Camden	2,518,438	6,670,341	7,378,745																																																																					
Grootvlei	2,992,859	2,445,162	2,416,293																																																																					
Komati	708,862	608,728	369,330																																																																					
Kusile	8,553,383	7,992,189	9,119,476																																																																					
Medupi	19,753,895	22,637,675	23,246,212																																																																					
Total	183,553,489	184,568,589	171,130,842																																																																					
Justification of choice of data or description of	Eskom is South Africa’s national electricity utility, responsible for generating electricity, procuring renewable electricity from																																																																							

measurement methods and procedures applied	Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 10
Purpose of data	To calculate baseline emissions, using Equation 4.
Comments	The generation-weighted average CO ₂ emissions per unit net electricity generation (t CO ₂ /MWh) of all generating power plants serving the system, not including low-cost/must-run power plants is used to determine the simple OM emission factor, using Equation 3 under Option A from section 48 in TOOL 07.

Data / Parameter	$EF_{EL,m,y}$			
Data unit	tCO ₂ /MWh			
Description	CO ₂ emission factor of power unit m in year y			
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (AEL) requirements.			
Value applied	As per Table 10			
	Plant Name	Emission factor 2023 (tCO ₂ /MWh)	Emission factor 2022 (tCO ₂ /MWh)	Emission factor 2021 (tCO ₂ /MWh)
	Arnot	1.17	1.16	1.14
	Duvha	1.11	1.13	1.13
	Hendrina	1.35	1.24	1.20
	Kendal	1.02	1.05	1.04
	Kriel	1.32	1.28	1.22
	Lethabo	1.00	0.97	0.95
	Matimba	1.02	1.04	0.99
	Majuba	1.19	1.13	1.09
	Matla	1.30	1.27	0.99
	Tutuka	1.18	1.23	1.22
	Camden	1.10	1.16	1.25
	Grootvlei	1.21	1.27	1.24
	Komati	1.01	1.03	1.06
	Kusile	1.01	1.30	1.09
	Medupi	0.88	0.92	0.91

Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The power plants m are shown in Table 10).
Purpose of data	To calculate baseline emissions, using Equation 5.
Comments	The data on fuel consumption and electricity generation is available for the selected m power plants, so Equation 4 under Option A1 from paragraph 49(a) in TOOL 07 is applied.

Data / Parameter	$FC_{i,m,y}$			
Data unit	Tonnes			
Description	Amount of fuel type i consumed by power unit m in year y			
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements.			
Value applied	Plant Name	Coal usage (t) 2021	Coal usage (t) 2022	Coal usage (t) 2023
	Arnot	5,663,126	5,186,766	4,293,316
	Duvha	6,581,000	5,732,195	5,428,210
	Hendrina	3,024,527	3,024,981	1,486,626
	Kendal	9,272,316	9,362,475	9,395,998
	Kriel	6,694,760	7,838,445	6,865,127
	Lethabo	15,158,029	16,277,939	15,646,686
	Matimba	12,938,911	13,430,751	13,179,578
	Majuba	12,773,092	11,924,693	9,831,151
	Matla	10,694,976	10,842,908	9,536,046
	Tutuka	6,697,281	5,510,214	4,697,013
	Camden	1,569,620	3,932,224	4,129,344
	Grootvlei	1,938,931	1,544,269	1,518,602
	Komati	402,289	335,833	197,954
	Kusile	5,047,912	4,783,403	5,035,930
	Medupi	9,763 379	11,351,788	11,181,866
	Plant Name	Fuel oil usage (t) 2023	Fuel oil usage (t) 2022	Fuel oil usage (t) 2021
	Arnot	45,179.7	37 ,792.2	36,433.7

	Duvha	71,179.4	39,550.9	34,435.1
	Hendrina	7,722.1	11,012.5	7,013.8
	Kendal	106,402.9	61,174.9	32,683.4
	Kriel	53,949.1	39, 516.2	47,877.9
	Lethabo	14,075.3	12,792.0	9,740.7
	Matimba	14,066.5	9, 944.3	13,465.6
	Majuba	140,169.2	149, 002.7	89,746,1
	Matla	15,923.0	14,520.0	9,179.0
	Tutuka	64,592.8	68,779.,3	124,521.3
	Camden	49,016.8	57,014.4	41 321,3
	Grootvlei	21,199.1	61,318.1	19 380,3
	Komati	1,773.0	1 332,0	2,214.0
	Kusile	10,351.0	534,043.2	32,165.0
	Medupi	23,246.2	22,637.7	19,753.9
	As per Table 11			
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The power plants m are shown in Table 11			
Purpose of data	To calculate baseline emissions, using Equation 5.			
Comments	The data on fuel consumption and electricity generation is available for the selected m power plants, so Equation 4 under Option A1 from paragraph 49(a) in TOOL 07 is applied.			

Data / Parameter	$NCV_{i,y}$		
Data unit	GJ/tonne		
Description	Net calorific value of fuel type i in year y		
Source of data	DFFE for coal [24] IPCC default factors for heavy fuel oil , [25]		
Value applied	For coal as per plant in Table 11		
	Plant Name	Net calorific value coal (GJ/t)	Net calorific value fuel oil (GJ/t)
	Arnot	21.43	40.40
	Duvha	20.39	40.40
	Hendrina	20.51	40.40
	Kendal	17.87	40.40

	Kriel	21.64	40.40
	Lethabo	14.53	40.40
	Matimba	19.15	40.40
	Majuba	19.58	40.40
	Matla	18.19	40.40
	Tutuka	19.83	40.40
	Camden	20.45	40.40
	Grootvlei	20.01	40.40
	Komati	19.72	40.40
	Kusile	19.42	40.40
	Medupi	19.42	40.40
Justification of choice of data or description of measurement methods and procedures applied	Tier 2 NCV (for South Africa) only available for coal. Tier 1 used for other NCVs		
Purpose of data	To calculate baseline emissions, using Equation 5.		
Comments	The data on fuel consumption and electricity generation is available for the selected m power plants, so Equation 4 under Option A1 from paragraph 49(a) in TOOL07 is applied.		

Data / Parameter	AEG_{total}
Data unit	GWh
Description	Total electricity generation of all power plants directly supplying the system in year y
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (AEL) requirements.
Value applied	184,165.63
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 10.
Purpose of data	To calculate baseline emissions
Comments	

Data / Parameter	$SET_5 \text{ units}$
Data unit	Dimensionless
Description	The set of five power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently, and that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements.
Value applied	Kusile (2017/08/30) Medupi (2015/08/22) Sere Wind (2015/03/01) Komati (2009/03/31) Grootvlei (2008/04/40)
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 12
Purpose of data	To calculate baseline emissions
Comments	This set of power units generate 19.2% of AEG_{total} in 2023.

Data / Parameter	$AEG_{set-5-units}$
Data unit	GWh
Description	Annual electricity generation of the set of five power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently, and that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements.
Value applied	35,365.31
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 12
Purpose of data	To calculate baseline emissions

Comments	This set of power units generated 19,2% of AEG_{total} in 2023.
Data / Parameter	$SET_{\geq 20\%}$
Data unit	Dimensionless
Description	The set of power units, excluding power units registered as CDM project activities, and that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [14] and monthly records published on Eskom's website [23] for each power station to fulfil the Atmospheric emission license (EAL) requirements.
Value applied	Kusile (2017/08/30) Medupi (2015/08/22) Ankerlig (2007/03/01) Gourikwa (2007/07/01) Grootvlei (2008/04/30) Komati (2009/03/31) Sere Wind (2015/03/01)
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 13
Purpose of data	To calculate baseline emissions
Comments	This set of power units generate 20.8% of AEG_{total} in 2023.
Data / Parameter	$AEG_{set \geq 20\%}$
Data unit	GWh
Description	Annual electricity generation of the set of power units, excluding power units registered as CDM project activities, and excluding those commissioned more than 10 years ago, and that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [14] and monthly records published on Eskom's website [23] for each power station to fulfil the Atmospheric emission license (EAL) requirements.
Value applied	38,379.02
Justification of choice of data or description of	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from

measurement methods and procedures applied	Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 13
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated 20,8% of AEG_{total} in 2023.

Data / Parameter	SET_{sample}
Data unit	Dimensionless
Description	The set of power units, (between $SET_{5\text{ units}}$ and $SET_{\geq 20\%}$) that comprises the larger annual electricity generation, and exclude power units that started to supply electricity to the grid more than 10 years ago
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [14] and monthly records published on Eskom's website [23] for each power station to fulfil the Atmospheric emission license (EAL) requirements.
Value applied	Kusile (2017/08/30) Medupi (2015/08/22) Sere Wind (2015/03/01)
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 14 and Table 13
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated 32,579,69 GWh in 2023, comprising 17,7% of AEG_{total} in 2023.

Data / Parameter	$AEG_{set\ sample}$
Data unit	GWh
Description	Annual electricity generation of the set of power units, excluding power units registered as CDM project activities, and excluding those commissioned more than 10 years ago, and that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [14] and monthly records published on Eskom's website [23] for each power station to fulfil the Atmospheric emission license (EAL) requirements.

Value applied	32,579.69
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 14 and Table 13
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated >20% of AEG_{total} in 2023.

Data / Parameter	$SET_{sample-CDM}$
Data unit	Dimensionless
Description	The set of power units, (between $SET_{5\text{ units}}$ and $SET_{\geq 20\%}$) that comprises the larger annual electricity generation, excludes power units that started producing electricity more than 10 years ago, and including power units registered as CDM project activities, that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements.
Value applied	Shown in Table 14.
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 14
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated 36 861,54 GWh in 2023, comprising 20,0% of AEG_{total}

Data / Parameter	$AEG_{SET-sample-CDM}$
Data unit	GWh
Description	Annual electricity generation of the set of power units, excluding power units registered as CDM project activities, and excluding those commissioned more than 10 years ago, and that comprise 20% of AEG_{total}

Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [11] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [14] and monthly and annual records published on Eskom's website [22] for each power station to fulfil the Atmospheric emission license (EAL)
Value applied	36,661.54
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 14
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated $\geq 20,0\%$ of AEG_{total} in 2023.
Data / Parameter	w_{OM}
Data unit	Percentage (%)
Description	Weighting of operating margin emissions factor
Source of data	TOOL 07 paragraph 86 (a)
Value applied	75%
Justification of choice of data or description of measurement methods and procedures applied	Default value provided in the CDM TOOL 07.
Purpose of data	To calculate baseline emissions, using Equation 7
Comments	Owing to the intermittent and non-dispatchable nature of solar power generation, the weighting of the operating margin emission factor is set at 75% for the first crediting period and for subsequent crediting periods.
Data / Parameter	w_{BM}
Data unit	Percentage (%)
Description	Weighting of build margin emissions factor
Source of data	TOOL 07 paragraph 86 (a)

Value applied	25%
Justification of choice of data or description of measurement methods and procedures applied	Default value provided in the CDM TOOL 07.
Purpose of data	To calculate baseline emissions, using Equation 7
Comments	Owing to the intermittent and non-dispatchable nature of solar power generation, the weighting of the build margin emission factor is set at 25% for the first crediting period and for subsequent crediting periods.

Data / Parameter	BE_y
Data unit	tCO ₂ e/yr
Description	Emission reductions in year y
Source of data	Quantified using Equation 1
Value applied	To be determined ex-post: 369,232
Justification of choice of data or description of measurement methods and procedures applied	Equation provided in ACM0002.
Purpose of data	To calculate baseline emissions, using Equation 1
Comments	No additional comments

5.2 Data and Parameters Monitored

The table below provides all data and parameters that will be monitored during the project crediting period. The values provided are used to quantify the estimated reductions and removals for the project crediting period in Section 4 above.

Data / Parameter	$EG_{facility,y}$
Data unit	MWh/Year

Description	Quantity of net electricity generation supplied by the project (i.e. PV, Wind and BESS facilities including the HPPC and VPPOC) plant to the grid in year y																		
Source of data	Electricity reading meters records																		
Description of measurement methods and procedures to be applied	Electricity generated is metered automatically, recorded electronically on the DCS and SCADA systems, recorded monthly, following the Plant's Standard Operating Procedures (SOPs), and checked against the Eskom meters (At Avondale site, meters are located at Eskom's Avondale switching station and Dassiesridge site, they are located at Eskom's Gringley switching station.)																		
Frequency of monitoring/recording	Continuous monitoring and monthly reporting.																		
Value applied	Ex ante values for the first crediting period: <table border="1"> <tr> <td>15-October-2025 to 31-December-2025</td><td>76,923</td></tr> <tr> <td>01-January-2026 to 31-December-2026</td><td>369,232</td></tr> <tr> <td>01-January-2027 to 31-December-2027</td><td>369,232</td></tr> <tr> <td>01-January-2028 to 31-December-2028</td><td>369,232</td></tr> <tr> <td>01-January-2029 to 31-December-2029</td><td>369,232</td></tr> <tr> <td>01-January-2030 to 31-December-2030</td><td>369,232</td></tr> <tr> <td>01-January-2031 to 31-December-2031</td><td>369,232</td></tr> <tr> <td>01-January-2032 to 14-October-2032</td><td>292 309</td></tr> <tr> <td>Total</td><td>2,584,626</td></tr> </table>	15-October-2025 to 31-December-2025	76,923	01-January-2026 to 31-December-2026	369,232	01-January-2027 to 31-December-2027	369,232	01-January-2028 to 31-December-2028	369,232	01-January-2029 to 31-December-2029	369,232	01-January-2030 to 31-December-2030	369,232	01-January-2031 to 31-December-2031	369,232	01-January-2032 to 14-October-2032	292 309	Total	2,584,626
15-October-2025 to 31-December-2025	76,923																		
01-January-2026 to 31-December-2026	369,232																		
01-January-2027 to 31-December-2027	369,232																		
01-January-2028 to 31-December-2028	369,232																		
01-January-2029 to 31-December-2029	369,232																		
01-January-2030 to 31-December-2030	369,232																		
01-January-2031 to 31-December-2031	369,232																		
01-January-2032 to 14-October-2032	292 309																		
Total	2,584,626																		
Monitoring equipment	Electricity meter, (to be confirmed during installation and commissioning), In access Distributed Control System (DCS), which will operate and run the Virtual Power Plant (VPP). Inaccess Distributed Control System (DCS) is used to monitor generated electricity internally as part of an overall Supervisory Control and Data Acquisition (SCADA) system. A primary meter, with an identical failsafe backup meter will be located at each site within the Eskom switching stations (Avondale and Gringley) on the project side of the grid tie in. These will be located at both the Avondale and Dassiesridge site. A check meter (with failsafe																		

	backup) will also be on site within the switching station on the national utility side of the grid tie in.
QA/QC procedures to be applied	Internal Operational QA/QC for meter reading via the SCADA system have been developed and will meet globally acceptable industry norms. Specifics of the QA/QC will be codified during the implementation and commission of the plant and included at verification. Calibration of the energy meter will be conducted by competent technicians under SANAS accreditation. The applicable national standard to be applied is SANS 474 [26], and will form part of planned maintenance schedules. This defines the required frequency of calibration in §4.7.4 for meters greater than 10MVA as being every 5 years and less than 10MVA as being every 10 years.
Purpose of data	Calculation of project emissions
Calculation method	Net electricity supplied to the grid by the project plant (i.e. PV, Wind and BESS facilities including the HPPC and VPPOC) is directly measured and reported, not calculated.
Comments	No additional comments

Data / Parameter	PE_y
Data unit	tCO ₂ /yr
Description	Project emissions in year y
Source of data	Quantified using Equation 8
Description of measurement methods and procedures to be applied	Quantified, not measured
Frequency of monitoring/recording	Monthly
Value applied	0
Monitoring equipment	Quantified, not monitored
QA/QC procedures to be applied	Describe the quality assurance and quality control (QA/QC) procedures to be applied, including the calibration procedures where applicable.
Purpose of data	Calculation of project emissions

Calculation method	Equation 8
Comments	For all renewable energy power generation project activities, emissions due to the use of fossil fuels for the backup generator can be neglected. This project does not involve geothermal power plants or water reservoir of hydro power plants. Therefore, project emissions for this renewable energy power generation project will be zero.

Data / Parameter	ER_y
Data unit	tCO ₂ /yr
Description	Emission reductions in year y
Source of data	Quantified using Equation 9
Description of measurement methods and procedures to be applied	Quantified, not measured
Frequency of monitoring/recording	Monthly
Value applied	To be calculated ex-post – Ex-ante value is 369,232 per annum
Monitoring equipment	Quantified, not monitored
QA/QC procedures to be applied	Describe the quality assurance and quality control (QA/QC) procedures to be applied, including the calibration procedures where applicable.
Purpose of data	Calculation of project emissions
Calculation method	Equation 9
Comments	The emission reductions determined at validation apply estimates of likely emission reductions from the proposed project activity during the crediting period, using available ex ante values. These will be revised during each verification using the actual data recorded (where applicable).

5.3 Monitoring Plan

Understanding that the project is still in design and implementation phase and has not been constructed and final equipment and instrumentation is still to be installed and implemented.

This is the best understanding at the time of validation and will be amended and corrected in an “as-built” version of the plant.

Methods for measuring, recording, storing, aggregating, collating and reporting on monitored data and parameters.

Electricity generated by the facility will be monitored internally using an InAccess Distributed Control System (DCS), as part of an overall Supervisory Control and Data Acquisition (SCADA) system which will be used for operational control and reporting as part of the VPP. Apart from reporting for this project, the data will also be used for billing the national utility for electricity supplied to the grid. All data and reports will be stored and backed up in accordance with the SCADA system design. A primary meter, with an identical failsafe backup meter will be located at each site within the Eskom switching stations (Avondale and Gringley) on the project side of the grid tie in. These will be located at both the Avondale and Dassiesridge site. These will form the core of the VPPOC monitoring and will form the basis for the emission reduction calculations. A check meter (with failsafe backup) will also be on site within the switching station on the national utility side of the grid tie in.

Local HPPC (Hybrid Power Plant Controller) will be installed on each power plant (Avondale and Dassiesridge site which actively controls the assets on each site in real time. The VPPOC (Virtual Power Plant Optimiser & Controller) is a sophisticated software system that aggregates and manages distributed energy resources to act as a single, virtual power plant. It optimizes the operation of these sources, such as solar panels, WEF, that feed the BESS, to improve grid stability, reduce costs, and enhance energy efficiency.

Measurement data will be managed by the project’s DCS. The measuring equipment will be installed in three steps:

- Calibration under standard temperature and pressure conditions
- Installation and electrical connection by a certified external contractor
- Commissioning and final test under project conditions by the internal instrumentation team

All data will be stored on a server on-site and also loaded onto a firewall protected cloud storage system. All measuring equipment will strictly comply with their specified maintenance plans and calibration certificates will be kept on record.

As per the applied methodology, electricity generation will be measured onsite at the outgoing substation, adjacent to the generating turbines. This will be monitored on a continuous basis, with a monthly report being aggregated and reported upon. A check meter, owned by the national utility

(Eskom) located at the Eskom Switching Stations which serves as the grid tie-in point. Meters at both locations will have a back-up meter installed as a failsafe.

The meters on site, will be calibrated in accordance with the manufacturer's specifications, and will form part of the planned maintenance schedule for both checking and calibration. Instrument accuracy checks and verification will be conducted by Competent Instrumentation technicians and records will be kept on site. Calibration of the energy meter will be conducted by competent technicians under SANAS accreditation. The applicable national standard to be applied is SANS 474 [26], and will form part of planned maintenance schedules. This defines the required frequency of calibration in §4.7.4 for meters greater than 10MVA as being every 5 years and less than 10MVA as being every 10 years.

Organizational structure, responsibilities and competencies of the personnel that will be carrying out monitoring activities.

Data collection, consolidation and results analysis will be undertaken by a dedicated data and instrumentation team who are adequately trained and competent. They will be responsible for the maintenance (planned and unplanned), checking and arranging calibration for plant equipment. They will also be responsible for maintaining records of calibrations and procurement of any instrumentation equipment that may be required for the operation of the plant. All instrument technicians will be suitably competent and records of their training and qualifications will be maintained by the Human Resources department. Only SANAS qualified organisations will conduct calibrations.

Procedures for internal auditing and QA/QC

Metered electricity production will be recorded by the plant on the DCS system. This will be checked against the Eskom meter at the independent switching stations. Any discrepancies will immediately trigger fault finding and corrective action procedures. There is also a backup meter both on site and at the switching stations which are used as an additional failsafe for QA/QC purposes. All reported electricity generation values will be checked by both project personnel and Eskom personnel prior to billing and payment.

APPENDIX 1: COMMERCIALY SENSITIVE INFORMATION

No commercially sensitive information is presented in this document

APPENDIX 2: BIBLIOGRAPHY

- [1] “Umoyilanga,” EDF Renewables. Accessed: Jan. 22, 2025. [Online]. Available: <https://edf-re.co.za/construction/umoyilanga/>
- [2] “Power Purchase Agreement Umoyilanga - Eskom.pdf.” Accessed: Oct. 24, 2024. [Online]. Available: https://anthesisllc.sharepoint.com/teams/External_-_Umoyilanga_Hybrid_RE_BESS_Project_365/Shared%20Documents/EDF%20project%20documents/Legal%20and%20Commercial/PPA%20-%20Umoyilanga%20-%20Executed%20version%20no%20Sch9.pdf?CT=1729779667163&OR=ItemsView&wdOrigin=OUTLOOKGROUPS.FILEBROWSER.DOCUMENTLIBRARY
- [3] “Umoyilanga (RF)(Pty) Ltd Nersa Generation licence.pdf.” Accessed: Oct. 24, 2024. [Online]. Available: [https://anthesisllc.sharepoint.com/teams/External_-_Umoyilanga_Hybrid_RE_BESS_Project_365/Shared%20Documents/EDF%20project%20documents/Legal%20and%20Commercial/Approval%20Letter%20for%20amendment%20of%20Umoyilanga%20\(RF\)\(Pty\)%20Ltd%20Generation%20licence.pdf?CT=1729780567817&OR=ItemsView&wdOrigin=OUTLOOKGROUPS.FILEBROWSER.DOCUMENTLIBRARY](https://anthesisllc.sharepoint.com/teams/External_-_Umoyilanga_Hybrid_RE_BESS_Project_365/Shared%20Documents/EDF%20project%20documents/Legal%20and%20Commercial/Approval%20Letter%20for%20amendment%20of%20Umoyilanga%20(RF)(Pty)%20Ltd%20Generation%20licence.pdf?CT=1729780567817&OR=ItemsView&wdOrigin=OUTLOOKGROUPS.FILEBROWSER.DOCUMENTLIBRARY)
- [4] “Avondale_1 - FULL Final EIAR and Annexures.pdf.”
- [5] “Final Basic Assessment Report_Dassiesridge BESS.pdf.”
- [6] “Umoyilanga,” EDF Renewables. Accessed: Oct. 24, 2024. [Online]. Available: <https://edf-re.co.za/construction/umoyilanga/>
- [7] F. Ruiters, “Dawid Kruiper Municipality integrated development plan for 2022/2027”.
- [8] “PROFILE: zf mgcawu district”.
- [9] “20210712 Umoyilanga - Avondale Final EMPr Approval - 2-618 (1).pdf.”
- [10] “Umo_132kV OHL final layout & EMPr approval- Dassiesridge.pdf.”
- [11] “2. Umoyilanga_Dassiesridge WEF_Final EMPr- Dassiesridge (1).pdf.”
- [12] “Avondale SPark - Final EMPr 2021-06-15.pdf.”
- [13] “FINAL Dassieridge WEF EIR.pdf.crdownload.”
- [14] “4.Eskom_integrated_report_2023.pdf.”
- [15] “south-africas-net-zero-transition.pdf.” Accessed: Sep. 06, 2024. [Online]. Available: <https://web-assets.bcg.com/79/c5/dc68d7e940d898730da40d74a511/south-africas-net-zero-transition.pdf>
- [16] “CDM Guidelines on the assessment of financial analysis v05.pdf.”
- [17] “South Africa Lending Rate.” Accessed: Oct. 10, 2024. [Online]. Available: <https://tradingeconomics.com/south-africa/lending-rate>
- [18] “South Africa Core Inflation Rate.” Accessed: Dec. 04, 2024. [Online]. Available: <https://tradingeconomics.com/south-africa/core-inflation-rate>
- [19] “Renewable Statistics - Eskom Data Portal.” Accessed: Oct. 11, 2024. [Online]. Available: <https://www.eskom.co.za/dataportal/renewables-performance/renewable-statistics/>

- [20] “The Country’s first Wind, Solar and Battery Hybrid Projects signed by DMRE & Eskom | South Africa Wind Energy Association.” Accessed: Oct. 11, 2024. [Online]. Available: <https://sawea.org.za/news/countrys-first-wind-solar-and-battery-hybrid-projects-signed-dmre-eskom>
- [21] “Historic_Electricity_Generation_per_Plant_Fossil-Fired.pdf.” Accessed: Aug. 28, 2024. [Online]. Available: https://www.eskom.co.za/dataportal/wp-content/uploads/2024/04/Historic_Electricity_Generation_per_Plant_Fossil-Fired.pdf
- [22] “Historic_Electricity_Generation_per_Plant_Fossil-Fired.pdf.” Accessed: Sep. 10, 2024. [Online]. Available: https://www.eskom.co.za/dataportal/wp-content/uploads/2024/04/Historic_Electricity_Generation_per_Plant_Fossil-Fired.pdf
- [23] “Atmospheric Emission License Reports - Eskom Data Portal.” Accessed: Jan. 22, 2025. [Online]. Available: <https://www.eskom.co.za/dataportal/emissions/ael/>
- [24] “1.DFFE Eskom emission factor report.pdf.”
- [25] “V2_1_Ch1_Introduction.pdf.” Accessed: Sep. 10, 2024. [Online]. Available: https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/2_Volume2/V2_1_Ch1_Introduction.pdf
- [26] “SANS 474:2006 NRS-057-Electricity-Metering.pdf.” Accessed: Sep. 02, 2024. [Online]. Available: <https://www.sseg.org.za/wp-content/uploads/2019/03/NRS-057-Electricity-Metering.pdf>